

PHOTOVOLTAIC ROOF MOUNT SYSTEM

12 MODULES-ROOF MOUNTED - 5.280 kW DC, 4.608 kW AC, 925 N GRANT ST, LAFAYETTE, OR 97127

PHOTOVOLTAIC SYSTEM SPECIFICATIONS:

SYSTEM SIZE:	5.280 KW DC 4.608 KW AC
MODULE TYPE & AMOUNT:	(12) ZNSHINESOLAR ZXM7-UHLD108-440/N
MODULE DIMENSIONS:	(L/W/H) 67.79"/44.64"/1.18"
INVERTER:	(06) AP SYSTEMS DS3-L [240V]
INTERCONNECTION METHOD:	LOAD BREAKER
AHJ:	CITY OF LAFAYETTE

GOVERNING CODES:

- 2023 OREGON RESIDENTIAL SPECIALTY CODE
- 2025 OREGON STRUCTURAL SPECIALTY CODE (OSSC)
- 2023 OREGON ELECTRICAL SPECIALTY CODE (NEC 2023)
- 2025 OREGON FIRE CODE

PHOTOVOLTAIC NOTES:

1. AN INVERTER OR AN AC MODULE IN AN INTERACTIVE PHOTOVOLTAIC SYSTEM SHALL AUTOMATIC ALL DE-ENERGIZE ITS OUTPUT TO THE CONNECTED ELECTRICAL PRODUCTION AND DISTRIBUTION NETWORK UPON LOSS OF VOLTAGE IN THAT SYSTEM AND SHALL REMAIN IN THAT STATE UNTIL THE ELECTRICAL PRODUCTION AND DISTRIBUTION NETWORK VOLTAGE HAS BEEN RESTORED. (NEC 690.361)
2. ALL EXTERIOR ELECTRICAL METALLIC TUBING(EMT) CONDUIT FITTING SHALL BE RAIN TIGHT THREAD-LESS COMPRESSION TYPE.
3. MODULES AND SUPPORT STRUCTURES SHALL BE GROUNDED
4. NAMEPLATES SHALL BE PROVIDED FOR ALL CIRCUITS IN THE SERVICE DISTRIBUTION AND POWER DISTRIBUTION SWITCH BOARDS, PANEL BOARDS, DISCONNECTING SWITCHES, TERMINAL CABINETS, ETC. ALL NAMEPLATES SHALL BE PERMANENTLY ATTACHED AND BE OF SUFFICIENT CAPACITY TO WITHSTAND THE WEATHER.
5. JUNCTION BOX/COMBINER BOX HAVE TO USE COMPRESSION TYPE STRAIN RELIEF POSITIONED FOR APPROPRIATE WATER RUN OFF.
6. CONDUIT RUNS SHALL BE PROVIDED WITH SUFFICIENT WEATHERPROOF PULL BOXES OF JUNCTION BOX/COMBINER BOXES PER APPROPRIATE NEC REQUIREMENTS.
7. SEE PROVIDED CUT SHEETS FOR ADDITIONAL EQUIPMENT SPECIFICATIONS
8. WIRING MATERIALS SHALL BE SUITABLE FOR THE SUN EXPOSURE AND WET LOCATIONS. FIELD APPLIED PROTECTIVE COATINGS ARE NOT ACCEPTABLE.
9. JUNCTION, PULL AND OUTLET BOXES LOCATED BEHIND MODULES SHALL BE SO INSTALLED THAT THE WIRING CONTAINED IN THEM CAN BE RENDERED ACCESSIBLE DIRECTLY OR BY DISPLACEMENT OF MODULE(S) SECURED BY REMOVABLE FASTENERS AND CONNECTED BY A FLEXIBLE WIRING SYSTEM. (NEC 690.34)
10. IN AN UNDERGROUND PHOTOVOLTAIC SYSTEM, THE POWER SOURCE SHALL BE LABELED WITH THE FOLLOWING WARNING AT EACH JUNCTION BOX, COMBINER BOX, DISCONNECT AND DEVICE WHERE THE UNGROUNDED CIRCUITS MAY BE EXPOSED DURING SERVICE : " WARNING - ELECTRIC SHOCK HAZARD. THE CURRENT CIRCUIT CONDUCTORS OF THIS PHOTOVOLTAIC POWER SYSTEM ARE UNGROUNDED BUT MAY BE ENERGIZED WITH THE RESPECT TO GROUND DUE TO LEAKAGE PATHS AND/OR GROUND (NECE 690.35(F))
11. ALL PHOTOVOLTAIC MODULES AND ASSOCIATED EQUIPMENT AND WIRING MATERIAL SHALL BE PROTECTED FROM ANY PHYSICAL DAMAGE.
12. ALL ELECTRICAL DEVICES AND UTILIZATION EQUIPMENT SHALL BE LISTED BY AN APPROVED TESTING AGENCY.
13. OUTDOOR EQUIPMENT SHALL BE AT LEAST NEMA 3R RATED.
14. ALL SPECIFIED WIRING IS BASED ON THE USE OF COPPER
15. CONTRACTOR SHALL OBTAIN ELECTRICAL PERMITS AND SHALL COORDINATE ALL INSPECTION, COMMISSIONING AND ACCEPTANCE WITH THE CLIENT, UTILITY CO. AND CITY INSPECTORS AS NEEDED
16. DRAWINGS ARE DIAGRAMMATIC ONLY, ROUTING OF RACEWAYS SHALL BE AT THE OPTION OF THE CONTRACTOR UNLESS OTHERWISE NOTED AND SHALL BE COORDINATED WITH OTHER TRADES.
17. IF DISTANCES OF CABLE RUNS ARE DIFFERENT THAN SHOWN, THE CONTRACTOR SHALL NOTIFY ELECTRICAL ENGINEER TO VALIDATE THE WIRE SIZE. FINAL DRAWINGS WILL BE RED-LINED AND UPDATED AS APPROPRIATE.
18. WHENEVER A DISCREPANCY IN QUANTITY OF EQUIPMENT, ARISES ON THE DRAWINGS OR SPECIFICATIONS, THE CONTRACTOR SHALL BE RESPONSIBLE FOR PROVIDING AND INSTALLING ALL MATERIAL AND SERVICES REQUIRED BY THE STRICTEST CONDITIONS NOTED ON THE DRAWINGS OR IN THE SPECIFICATIONS TO ENDURE COMPLETE COMPLIANCE AND LONGEVITY OF THE OPERABLE SYSTEM REQUIRED BY THE ARCHITECT/ENGINEER.
19. ALL BROCHURES, OPERATION MANUALS, CATALOGS, SHOP DRAWINGS, ETC. SHALL BE HANDED OVER TO THE OWNER'S REPRESENTATIVE AT THE COMPLETION OR WORK.
20. ALL WIRING CONCEALED IN WALL AND CEILING SPACES SHALL BE IN METAL CONDUIT.
21. THE SEISMIC BRACING AND ANCHORAGE OF ELECTRICAL CONDUITS SHALL BE IN ACCORDANCE

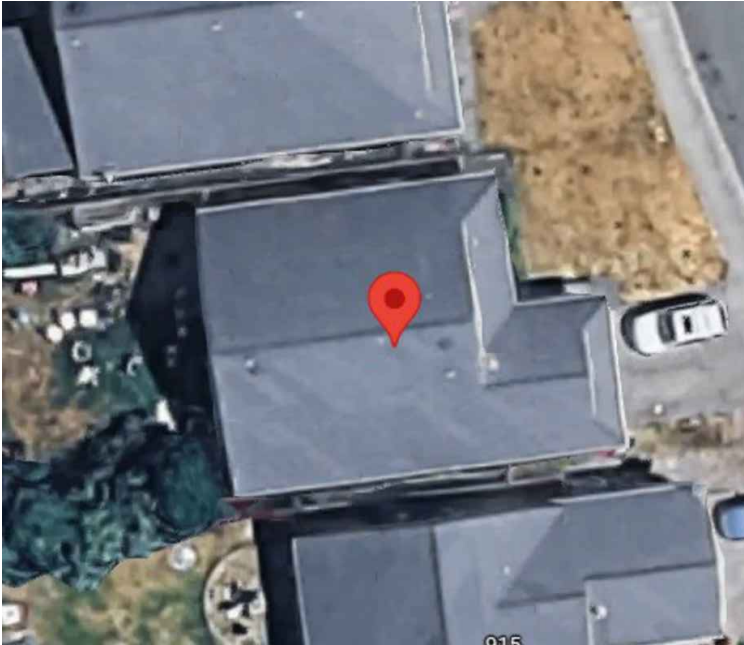
SITE DETAIL:

UTILITY:	PGE
MSP:	EXISTING 200A MSP WITH 200A MAIN BREAKER
ROOF MATERIAL:	COMPOSITE SHINGLE
ASCE 7-10 WINDSPEEDS:	(3 SEC GUST IN MPH)
WIND SPEED AND EXPOSURE:	110 MPH, C
ROOF SNOW LOAD:	20 PSF
DEAD LOAD FOR ROOF-MOUNTED PANELS ATTACHMENTS:	2.77 PSF
RACKING:	CHIKO RACKING SYSTEM FOR COMP ROOF
ARRAY AREA:	21.01 SQFT x 12 = 252.1 SQFT

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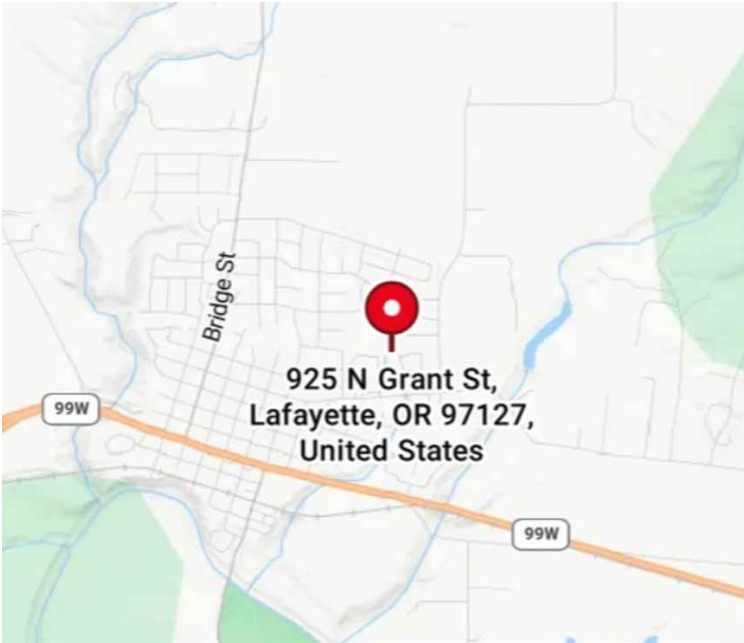


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SATELLITE VIEW

PV 0.0

SCALE: NTS



3

VICINITY MAP

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1

GENERAL NOTES AND PROJECT DATA

PV 0.0



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INFINITY HOME SOLUTIONS

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REVISIONS

Description	Date	Rev
Initial Design	6/16/26	00

Signature with Seal

Project Name &
Address

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GENERAL NOTES
AND PROJECT
DATA

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PHOTOVOLTAIC ROOF MOUNT SYSTEM

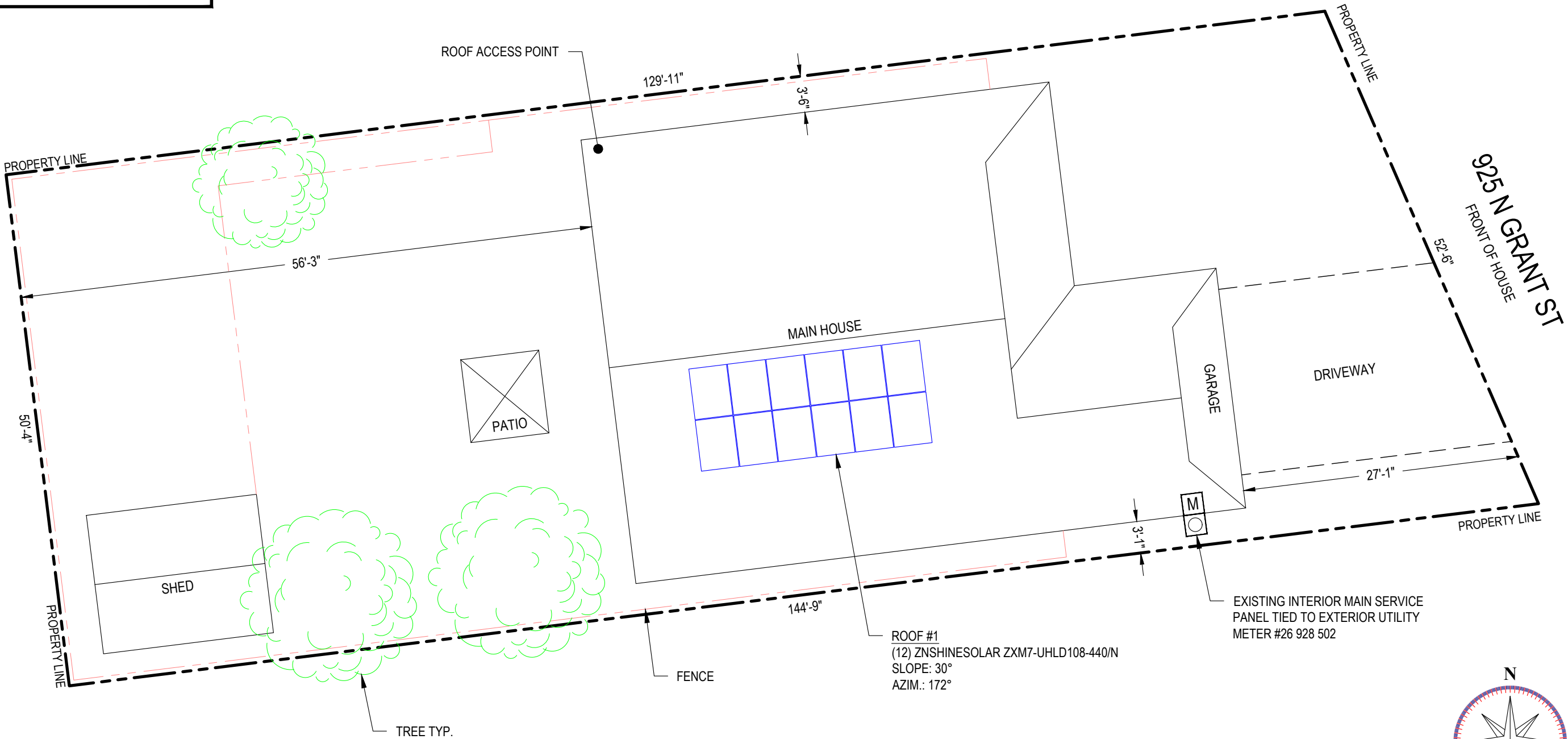
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AHJ:	CITY OF LAFAYETTE

NEW PV AC DISCONNECT VISIBLE,
LOCKABLE, LABELED DISCONNECT
WITHIN 10' OF UTILITY METER

● ROOF ACCESS POINT
ROOF ACCESS POINT SHALL NOT BE LOCATED IN AREAS THAT REQUIRE THE PLACEMENT OF GROUND LADDERS OVER OPENINGS SUCH AS WINDOWS OR DOORS, AND LOCATED AT STRONG POINTS OF BUILDING CONSTRUCTION IN LOCATIONS WHERE THE ACCESS POINT DOES NOT CONFLICT WITH OVERHEAD OBSTRUCTIONS SUCH AS TREE LIMBS, WIRES OR SIGNS.





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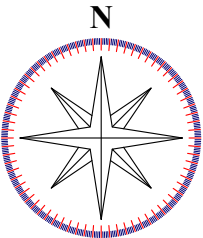
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PLOT PLAN

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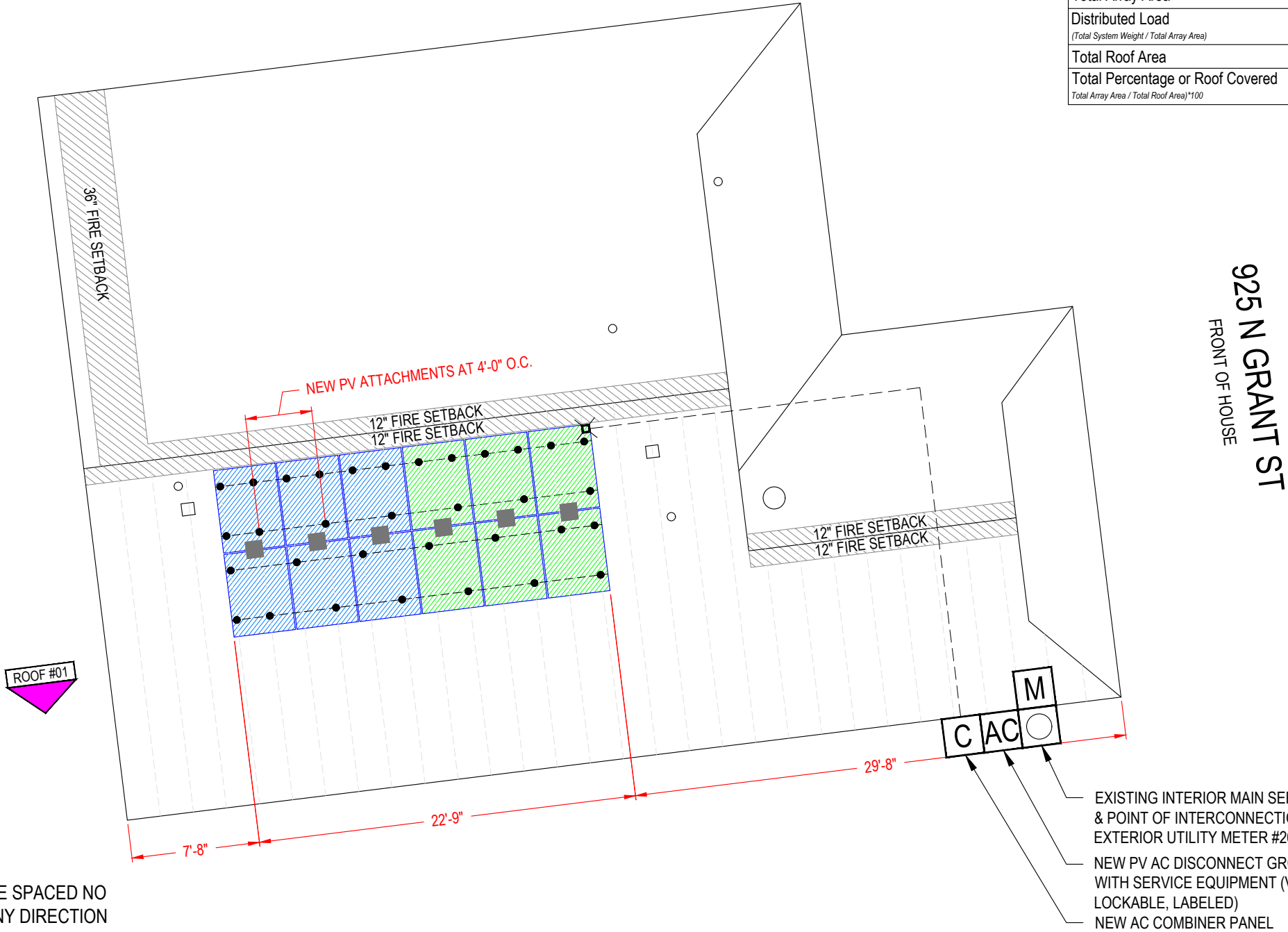
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AHJ: CITY OF LAFAYETTE

VISIBLE, LOCKABLE, LABELED DISCONNECT
WITHIN 10' OF UTILITY METER

NOTE :
ATTIC RUN - YES
ATTIC FAN - NO
SHUTDOWN - NO

BILL OF MATERIALS		
NUMBER OF MODULES	12	ZNSHINESOLAR ZXM7-UHLD108-440/N
NUMBER OF INVERTER	06	AP SYSTEMS DS3-L [240V]
COMBINER PANEL	1	EATON BR STYLE 1-INCH LOAD CENTER 816L125RP
AC DISCONNECT	1	30A NON-FUSIBLE AC DISCONNECT, 240V
NUMBER OF ATTACHMENTS	33	CHIKO AL ROOF HOOK #167 CK-FTS-167RT2
RAILS	7	CHIKO 518 RAIL 4200MM
RAIL SPLICE	4	SPLICE KIT
MID CLAMPS	20	MID CLAMPS / UFO
END CLAMPS	8	END CLAMPS / STOPPER SLEEVE
GROUNDING LUG	4	GROUNDING LUG

MODULE, ARRAY WEIGHT (LOAD CALC'S)		
Number of Modules	12	
Module Weight	54.01	LBS
Total Module (Array) Weight	648.12	LBS
Number of Attachment point	33	
Mounting System Weight (Per Module)	1.5	LBS
Mounting System Weight	49.50	LBS
Total System Weight (Module Weight + Mounting System Weight)	697.62	LBS
Weight at Each Attachment Point (Array Weight / Number of Attachment Point)	19.64	LBS
Module Area (67.79"x44.64")	21.01	SqFt
Total Array Area	252.18	SqFt
Distributed Load (Total System Weight / Total Array Area)	2.77	Per SqFt
Total Roof Area	2344	SqFt
Total Percentage or Roof Covered Total Array Area / Total Roof Area)*100	10.76%	



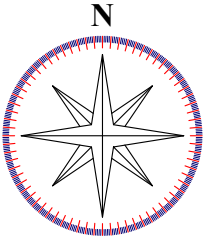
NOTE :-
ROOF ATTACHMENTS SHALL BE SPACED NO GREATER THAN 24 IN. OC IN ANY DIRECTION WHERE LOCATED WITHIN 3FT. OF A ROOF EDGE, HIP, EAVE OR RIDGE.

SYSTEM LEGEND

- EXISTING INTERIOR MAIN SERVICE PANEL & POINT OF INTERCONNECTION. TIED TO EXTERIOR UTILITY METER #26 928 502.
- NEW VISIBLE, LOCKABLE, LABELED DISCONNECT LOCATED WITHIN 10' FROM THE UTILITY METER
- NEW DEDICATED PV SYSTEM COMBINER PANEL.
- 12 NEW ZNSHINESOLAR ZXM7-UHLD108-440/N MODULES WITH NEW (06) AP SYSTEMS DS3-L [240V] INVERTERS.
- = FIRE PATHWAY
- = ROOF OBSTRUCTIONS
- = ATTACHMENT POINTS
- = RAFTER
- = RACKING SYSTEM
- = ATTIC RUN
- = CONDUIT ATTIC RUN JUNCTION BOX

ROOF SECTIONS

ROOF #01
MODULE - 12
SLOPE - 30°
AZIMUTH - 172°
MATERIAL - COMPOSITE SHINGLE
TRUSS SIZE & SPACING - 2"x4" @ 24" O.C.



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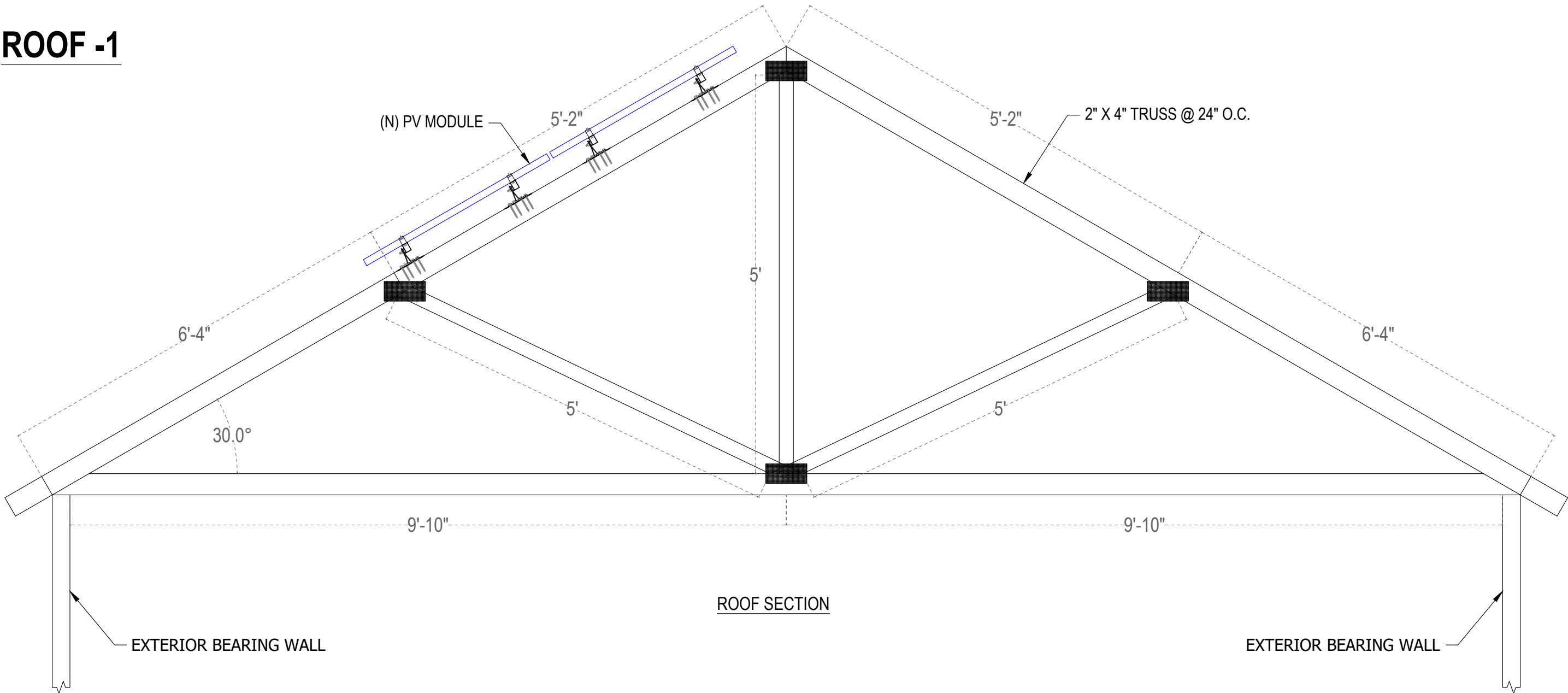
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SITE PLAN

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ROOF -1



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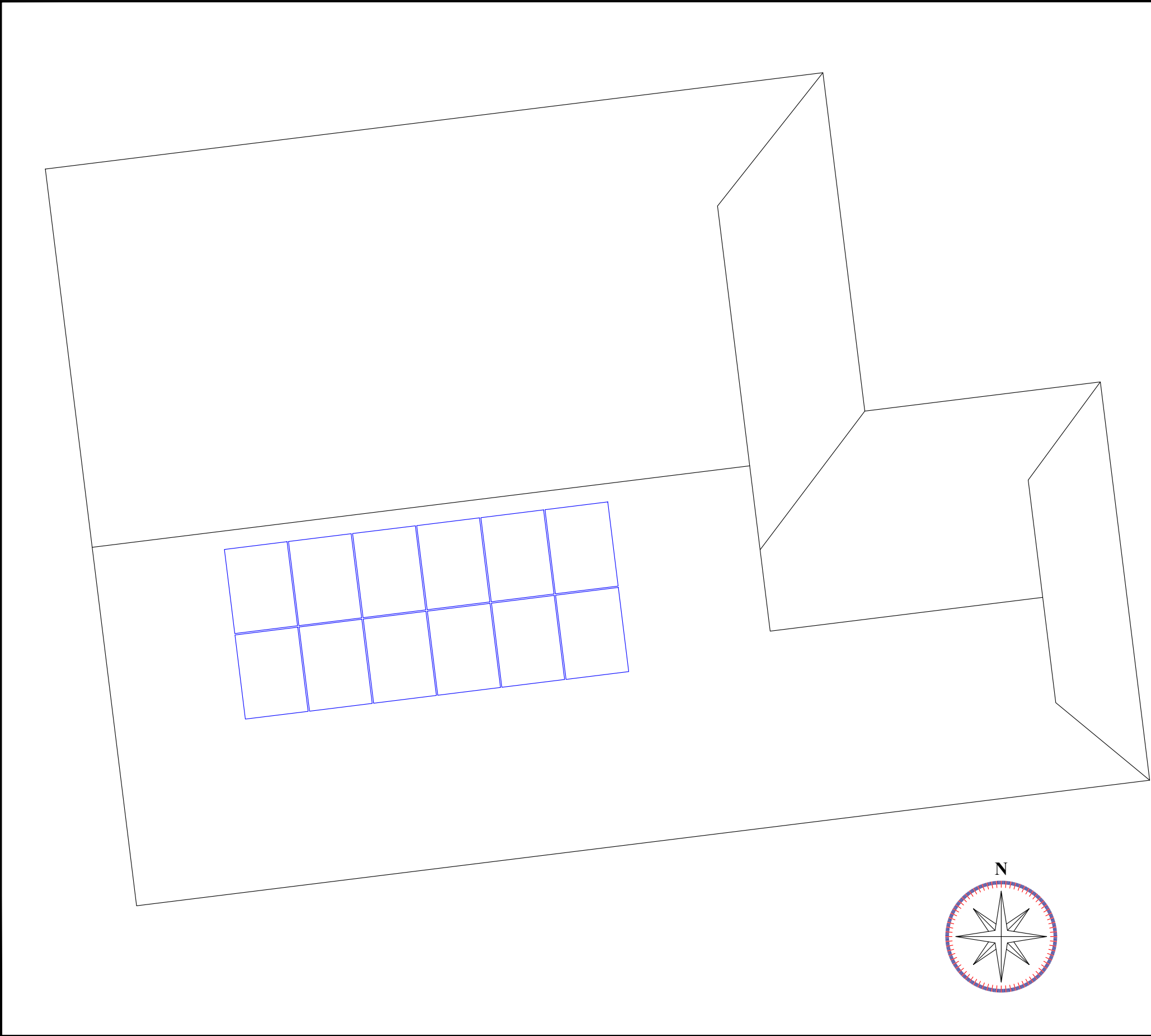
ROOF SECTION

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DTU/ECU:


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INVERTER MAP

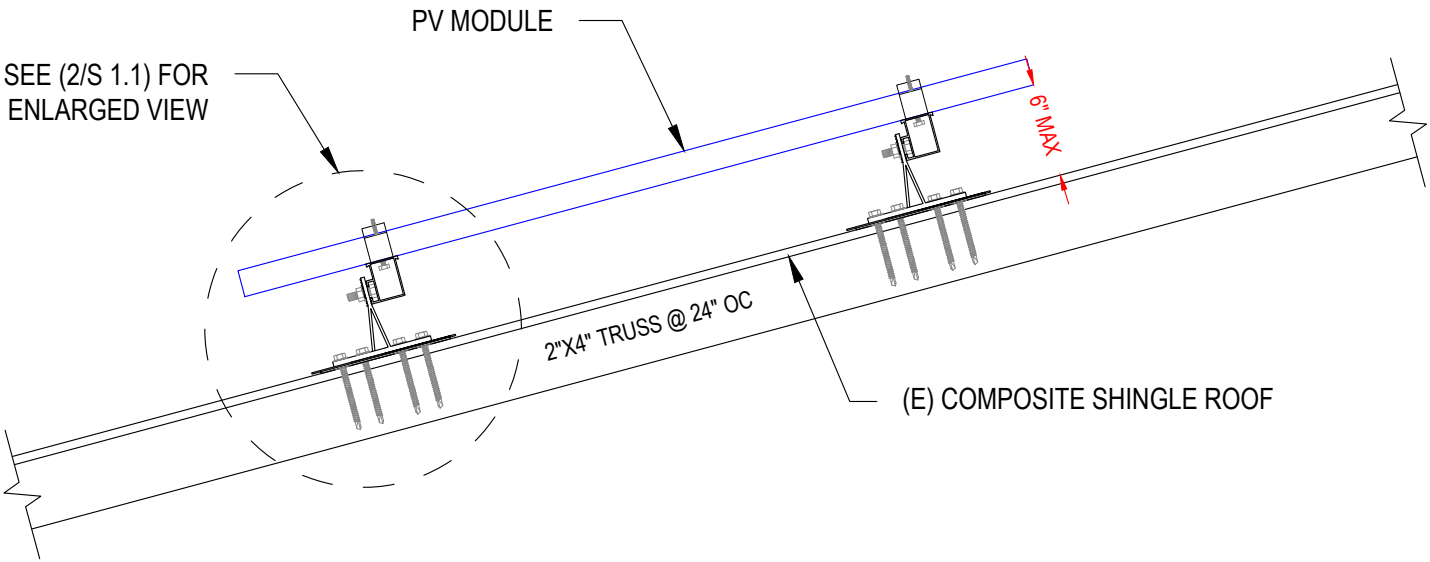
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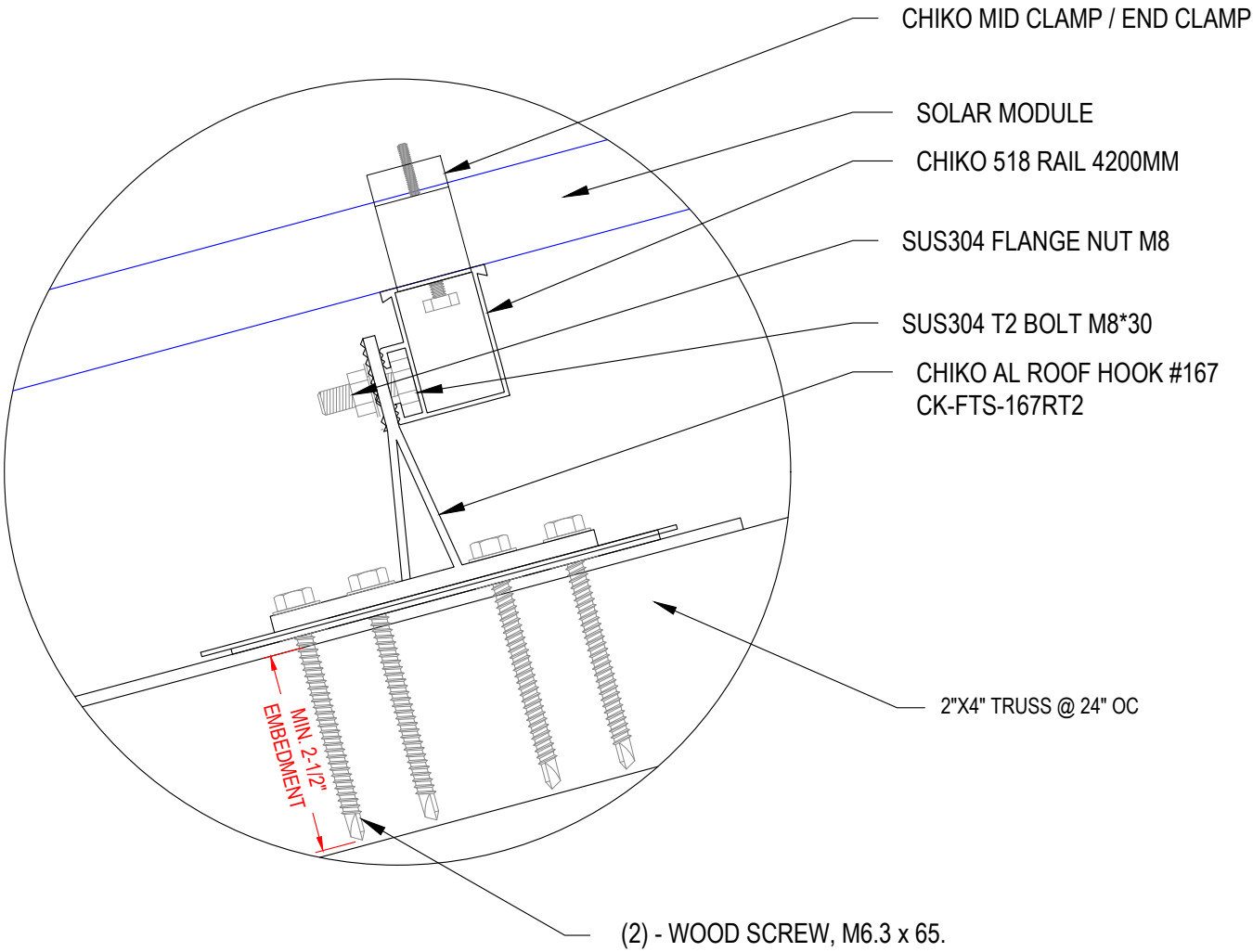
GENERAL STRUCTURAL NOTES:

THE SOLAR PANELS ARE TO BE MOUNTED TO THE ROOF FRAMING USING THE CHIKO RACKING SYSTEM WITH CHIKO AL ROOF HOOK #167 CK-FTS-167RT2 ATTACHMENTS. THE MOUNTING FEET ARE TO BE SPACED AS SHOWN IN THE DETAILS, AND MUST BE STAGGERED TO ADJACENT FRAMING MEMBERS TO SPREAD OUT THE ADDITIONAL LOAD.

- 1. THE PROPOSED PV SYSTEM ADDS 2.77 PSF TO THE ROOF FRAMING SYSTEM.
- 2. ROOF LIVE LOAD = 20 PSF TYPICAL, 0 PSF UNDER NEW PV SYSTEM.
- 3. GROUND SNOW LOAD = 20 PSF
- 4. WIND SPEED = 110 MPH
- 5. EXPOSURE CATEGORY = C
- 6. RISK CATEGORY = II



1 ATTACHMENT DETAIL (SIDE VIEW)
S 1.1 SCALE: NTS



NOTE: - A PANELS WILL NOT MOUNT HIGHER THAN 12 INCHES ABOVE THE SURFACE OF THE ROOF TOP WHICH THEY ARE MOUNTED.

2 ATTACHMENT DETAIL ENLARGED VIEW
S 1.1 SCALE: NTS



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MOUNT DETAIL

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MICROINVERTERS SPECIFICATION	
MICROINVERTER MAKE	AP SYSTEMS
MICROINVERTER MODEL NO.	DS3-L
MAXIMUM AC OUTPUT	768 W
MAX INPUT VOLTAGE	60 V
MAX AC CURRENT	3.20 A
MAX OUTPUT VOLTAGE	240 V

VISIBLE, LOCKABLE, LABELED DISCONNECT
WITHIN 10' OF UTILITY METER

NOTE :
ATTIC RUN - YES
SHUTDOWN - NO

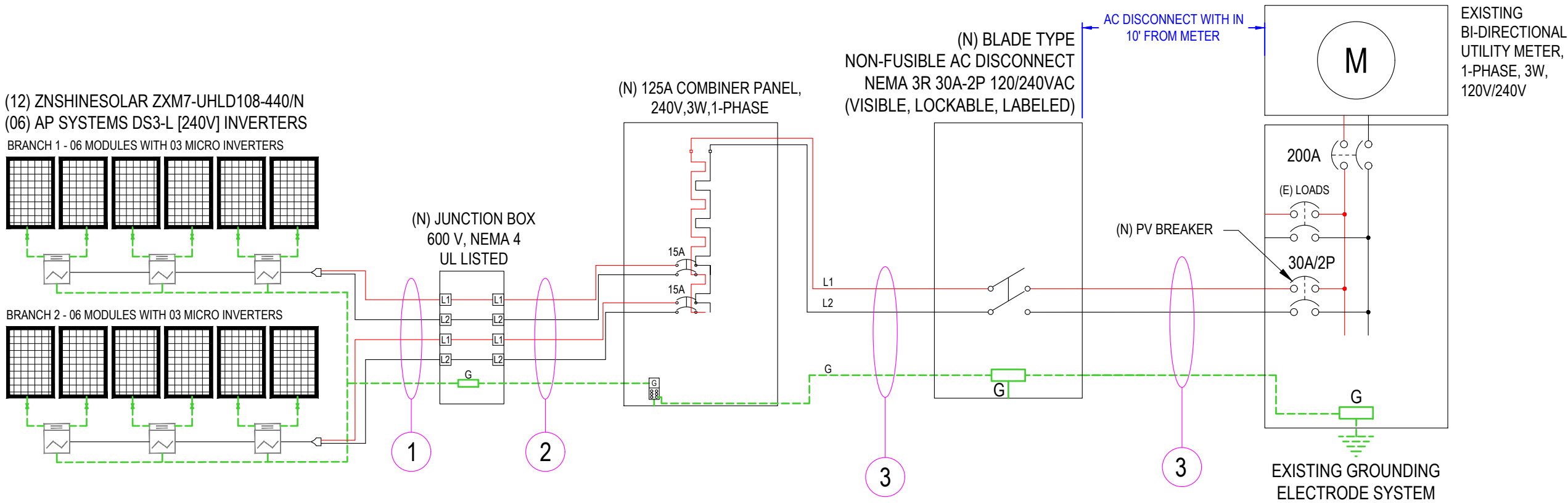
PV MODULE RATING @ STC	
MANUFACTURER	ZNSHINESOLAR ZXM7-UHLD108-440/N
MAX. POWER-POINT CURRENT (IMP)	13.63 AMPS
MAX. POWER-POINT VOLTAGE (VMP)	32.30 VOLTS
OPEN-CIRCUIT VOLTAGE (VOC)	38.90 VOLTS
SHORT-CIRCUIT CURRENT (ISC)	14.39 AMPS
NOM. MAX. POWER AT STC (PMAX)	440 WATT
MAX. SYSTEM VOLTAGE	1500V
VOC TEMPERATURE COEFFICIENT	-0.25 %/°C

MODULE: (12) ZNSHINESOLAR ZXM7-UHLD108-440/N
INVERTER: (06) AP SYSTEMS DS3-L [240V]

Rooftop conductor ampacities designed in compliance with art. 690.8, Tables 310.15(B)(1), 310.15(C)(1), 310.15(B)(2), 310.16, Chapter 9 Table 4, 5, & 9. Location specific temperature obtained from ASHRAE data tables	
RECORD LOW TEMP	-8°C
AMBIENT TEMP (HIGH TEMP 2%)	32°C
CONDUIT HEIGHT	7/8"
ROOF TOP TEMP	54°C
CONDUCTOR TEMPERATURE RATE	90°C

120% RULE	
BUS BAR RATING X 120%) - MAIN BREAKER RATING = MAX. PV OCPD	
(200A x 120%) - 200 = 40A	
THIS PANEL IS FED BY MULTIPLE SOURCES (UTILITY AND SOLAR)	
AC OUTPUT CURRENT	19A
NOMINAL AC VOLTAGE	240V

UTILITY SERVICE: UNDERGROUND
POINT OF INTERCONNECT, LOAD BREAKER
EXISTING 240V/200A BUS BAR RATING,
MAIN SERVICE PANEL, SINGLE PHASE,
WITH A 200A MAIN BREAKER
UTILITY COMPANY - PGE
UTILITY METER# 26 928 502



WIRE TAG #	MAX PARALLEL DEVICES		C.C RATING		PV CURR. MULT		TOTAL CURR	CONT. CURR X 125%	WIRE SIZE\TYPE\AMP.	WIRE OCP	#C.C.C.	CONDUIT	TEMP DE-RATE:	CONDUIT FILL:		WIRE AMP:	DERATED AMPACITY	MAX. CONT. CURRENT	GND	LENGTH FT. MAX	VOLTAGE DROP(%)		
①	03	x	3.20	x	N/A	=	9.60A	12.00A	#10\ THWN \ 40A @90°C	15A	2	TRUNK CABLE IN AIR	0.96	x	1.00	x	40A	38.40A	9.60A	#6 AWG	20	0.23	0.93<2
②	03	x	3.20	x	N/A	=	9.60A	12.00A	#10\ THWN \ 40A @90°C	15A	4	3/4" FMC / NMC IN AIR	0.96	x	0.80	x	40A	30.72A	9.60A	#10 AWG	40	0.57	
③	06	x	3.20	x	N/A	=	19.20A	24.00A	#10\ THWN \ 35A @75°C	30A	2	3/4" EMT	0.94	x	1.00	x	35A	32.90A	19.20A	#10 AWG	10	0.17	



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DIAGRAM

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SITE NOTES:

- 1. A LADDER WILL BE IN PLACE FOR INSPECTION IN COMPLIANCE WITH OSHA REGULATIONS.
- 2. THE PV MODULES ARE CONSIDERED NON-COMBUSTIBLE AND THIS SYSTEM IS A UTILITY INTERACTIVE SYSTEM WITH NO STORAGE BATTERIES.
- 3. THE SOLAR PV INSTALLATION WILL NOT OBSTRUCT ANY PLUMBING, MECHANICAL, OR BUILDING ROOF VENTS.
- 4. PROPER ACCESS AND WORKING CLEARANCE AROUND EXISTING AND PROPOSED ELECTRICAL EQUIPMENT WILL BE PROVIDED AS PER SECTION NEC 110.26.
- 5. ROOF COVERINGS SHALL BE DESIGNED, INSTALLED, AND MAINTAINED IN ACCORDANCE WITH THIS CODE AND THE APPROVED MANUFACTURER'S INSTRUCTIONS SUCH THAT THE ROOF COVERING SERVES TO PROTECT THE BUILDING OR STRUCTURE.

EQUIPMENT LOCATIONS:

- 1. ALL EQUIPMENT SHALL MEET MINIMUM SETBACKS AS REQUIRED BY NEC 110.26.
- 2. WIRING SYSTEMS INSTALLED IN DIRECT SUNLIGHT MUST BE RATED FOR EXPECTED OPERATING TEMPERATURE AS SPECIFIED BY NEC 690.31 (A),(C) AND NEC TABLES 310.15 (B)(1) AND 310.15 (B)(2).
- 3. JUNCTION AND PULL BOXES PERMITTED INSTALLED UNDER PV MODULES ACCORDING TO NEC 690.34.
- 4. ADDITIONAL AC DISCONNECT(S) SHALL BE PROVIDED WHERE THE INVERTER IS NOT WITHIN SIGHT OF THE AC SERVICING DISCONNECT. 2.2.6 ALL EQUIPMENT SHALL BE INSTALLED ACCESSIBLE TO QUALIFIED PERSONNEL ACCORDING TO NEC APPLICABLE CODES.
- 5. ALL COMPONENTS ARE LISTED FOR THEIR PURPOSE AND RATED FOR OUTDOOR USAGE WHEN APPROPRIATE.

STRUCTURAL NOTES:

- 1. RACKING SYSTEM & PV ARRAY WILL BE INSTALLED ACCORDING TO CODE-COMPLIANT INSTALLATION MANUAL. TOP CLAMPS REQUIRE A DESIGNATED SPACE BETWEEN MODULES, AND RAILS MUST ALSO EXTEND A MINIMUM DISTANCE BEYOND EITHER EDGE OF THE ARRAY / SUBARRAY, ACCORDING TO RAI MANUFACTURER'S INSTRUCTIONS.
- 2. JUNCTION BOX WILL BE INSTALLED PER MANUFACTURERS' SPECIFICATIONS. IF ROOF-PENETRATING TYPE, IT SHALL BE FLASHED & SEALED PER LOCAL REQUIREMENTS.
- 3. ROOFTOP PENETRATIONS FOR PV RACEWAY WILL BE COMPLETED AND SEALED W/ APPROVED CHEMICAL SEALANT PER CODE BY A LICENSED CONTRACTOR.
- 4. ALL PV RELATED ROOF ATTACHMENTS TO BE SPACED NO GREATER THAN THE SPAN DISTANCE SPECIFIED BY THE RACKING MANUFACTURER. 2.3.6 WHEN POSSIBLE, ALL PV RELATED RACKING ATTACHMENTS WILL BE STAGGERED AMONGST THE ROOF FRAMING MEMBERS.

WIRING & CONDUIT NOTES:

- 1. ALL CONDUIT AND WIRE WILL BE LISTED AND APPROVED FOR THEIR PURPOSE. CONDUIT AND WIRE SPECIFICATIONS ARE BASED ON MINIMUM CODE REQUIREMENTS AND ARE NOT MEANT TO LIMIT UP-SIZING.
- 2. CONDUCTORS SIZED ACCORDING TO NEC 690.8, NEC 690.7.
- 3. VOLTAGE DROP LIMITED TO 2%.
- 4. DC WIRING LIMITED TO MODULE FOOTPRINT. MICRO INVERTER WIRING SYSTEMS SHALL BE LOCATED AND SECURED UNDER THE ARRAY W/ SUITABLE WIRING CLIPS.
- 5. AC CONDUCTORS COLORED OR MARKED AS FOLLOWS: PHASE A OR L1- BLACK PHASE B OR L2- RED, OR OTHER CONVENTION IF THREE PHASE PHASE C OR L3- BLUE, YELLOW, ORANGE**, OR OTHER CONVENTION NEUTRAL- WHITE OR GREY IN 4-WIRE DELTA CONNECTED SYSTEMS THE PHASE WITH HIGHER VOLTAGE TO BE MARKED ORANGE [NEC 110.15].

GROUNDING NOTES:

- 1. GROUNDING SYSTEM COMPONENTS SHALL BE LISTED FOR THEIR PURPOSE, AND GROUNDING DEVICES EXPOSED TO THE ELEMENTS SHALL BE RATED FOR SUCH USE.
- 2. PV EQUIPMENT SHALL BE GROUNDED ACCORDING TO NEC 690.43 AND MINIMUM NEC TABLE 250.122.
- 3. METAL PARTS OF MODULE FRAMES, MODULE RACKING, AND ENCLOSURES CONSIDERED GROUNDED IN ACCORD WITH 250.134 AND 250.136.
- 4. EQUIPMENT GROUNDING CONDUCTORS SHALL BE SIZED ACCORDING TO NEC 690.45 AND MICRO INVERTER MANUFACTURERS' INSTRUCTIONS.
- 5. EACH MODULE WILL BE GROUNDED USING WEEB GROUNDING CLIPS AS SHOWN IN MANUFACTURER DOCUMENTATION AND APPROVED BY THE AHJ. IF WEEBS ARE NOT USED, MODULE GROUNDING LUGS MUST BE INSTALLED AT THE SPECIFIED GROUNDING LUG HOLES PER THE MANUFACTURERS' INSTALLATION REQUIREMENTS.
- 6. THE GROUNDING CONNECTION TO A MODULE SHALL BE ARRANGED SUCH THAT THE REMOVAL OF A MODULE DOES NOT INTERRUPT A GROUNDING CONDUCTOR TO ANOTHER MODULE.
- 7. GROUNDING AND BONDING CONDUCTORS, IF INSULATED, SHALL BE COLORED GREEN OR MARKED GREEN IF #4 AWG OR LARGER [NEC 250.119]
- 8. THE GROUNDING ELECTRODE SYSTEM COMPLIES WITH NEC 690.47 AND NEC 250.50 THROUGH 250.106. IF EXISTING SYSTEM IS INACCESSIBLE, OR INADEQUATE, A GROUNDING ELECTRODE SYSTEM PROVIDED ACCORDING TO NEC 250, NEC 690.47 AND AHJ.
- 9. GROUND-FAULT DETECTION SHALL COMPLY WITH NEC 690.41(B)(1) AND (2) TO REDUCE FIRE HAZARDS

DISCONNECTION AND OVER-CURRENT PROTECTION NOTES:

- 1. DISCONNECTING SWITCHES SHALL BE WIRED SUCH THAT WHEN THE SWITCH IS OPENED THE CONDUCTORS REMAINING ENERGIZED ARE CONNECTED TO THE TERMINALS MARKED "LINE SIDE" (TYPICALLY THE UPPER TERMINALS).
- 2. DISCONNECTS TO BE ACCESSIBLE TO QUALIFIED UTILITY PERSONNEL, BE LOCKABLE, AND BE A VISIBLE-BREAK SWITCH
- 3. PV SYSTEM CIRCUITS INSTALLED ON OR IN BUILDINGS SHALL INCLUDE A RAPID SHUTDOWN FUNCTION TO REDUCE SHOCK HAZARD FOR EMERGENCY RESPONDERS IN ACCORDANCE WITH 690.12(A) THROUGH (D).
- 4. ALL OCPD RATINGS AND TYPES SPECIFIED ACCORDING TO NEC 690.8, 690.9, AND 240.
- 5. MICRO INVERTER BRANCHES CONNECTED TO A SINGLE BREAKER OR GROUPED FUSES IN ACCORDANCE WITH NEC 110.3(B).
- 6. IF REQUIRED BY AHJ, SYSTEM WILL INCLUDE ARC-FAULT CIRCUIT PROTECTION ACCORDING TO NEC 690.11 AND UL1699B.

INTERCONNECTION NOTES:

- 1. LOAD-SIDE INTERCONNECTION SHALL BE IN ACCORDANCE WITH [NEC 705.12]
- 2. THE SUM OF THE UTILITY OCPD AND INVERTER CONTINUOUS OUTPUT MAY NOT EXCEED 120% OF BUS BAR RATING [NEC 705.12(B)(3)(2)].
- 3. THE SUM OF 125 PERCENT OF THE POWER SOURCE(S) OUTPUT CIRCUIT CURRENT AND THE RATING OF THE OVERCURRENT DEVICE PROTECTING THE BUS BAR SHALL NOT EXCEED 120 PERCENT OF THE AMPACITY OF THE BUS BAR, PV DEDICATED BACKFEED BREAKERS MUST BE LOCATED OPPOSITE END OF THE BUS FROM THE UTILITY SOURCE OCPD [NEC 705.12(B)(3)(1)].
- 4. AT MULTIPLE ELECTRIC POWER SOURCES OUTPUT COMBINER PANEL, TOTAL RATING OF ALL OVERCURRENT DEVICES SHALL NOT EXCEED AMPACITY OF BUSBAR. HOWEVER, THE COMBINED OVERCURRENT DEVICE MAY BE EXCLUDED ACCORDING TO NEC 705.12 (B)(3)(3).
- 5. FEEDER TAP INTERCONNECTION (LOADSIDE) ACCORDING TO NEC 705.12 (B)(1)
- 6. SUPPLY SIDE TAP INTERCONNECTION ACCORDING TO NEC 705.11 WITH SERVICE ENTRANCE CONDUCTORS IN ACCORDANCE WITH NEC 230.42 2.7.8BACKFEEDING BREAKER FOR ELECTRIC POWER SOURCES OUTPUT IS EXEMPT FROM ADDITIONAL FASTENING [NEC 705.12 (E)].



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VANCOUVER WA 98661
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REVISIONS		
Description	Date	Rev
Initial Design	6/16/26	00

Signature with Seal

Project Name & Address

KEVIN GONZALEZ RESIDENCE

925 N GRANT ST
LAFAYETTE, OR 97127
PHONE:- N/A
EMAIL:- N/A

Sheet Name
NOTES

Sheet Size
ANSI B
11" X 17"

Sheet Number
E 1.2

!

WARNING

ELECTRICAL SHOCK HAZARD

TERMINALS ON BOTH LINE AND LOAD SIDES MAY BE ENERGIZED IN THE OPEN POSITION

LABEL LOCATION:
WHERE ALL TERMINALS OF THE DISCONNECTING MEANS MAY BE ENERGIZED IN THE OPEN POSITION, A WARNING SIGN SHALL BE MOUNTED ON OR ADJACENT TO THE DISCONNECTING MEANS.
PER CODE(S): NEC 2020:
NEC 706.15(C)(4) and NEC 690.13(B)

!

WARNING

THIS EQUIPMENT FED BY MULTIPLE SOURCES, TOTAL RATING OF ALL OVERCURRENT DEVICES, EXCLUDING MAIN SUPPLY OVERCURRENT DEVICE, SHALL NOT EXCEED AMPACITY OF BUSBAR.

LABEL LOCATION:
PERMANENT WARNING LABELS SHALL BE APPLIED TO DISTRIBUTION EQUIPMENT, AC DISCONNECT
NEC: 2020
PER CODE: 705.12(B)(3)(3)

PHOTOVOLTAIC

AC DISCONNECT

LABEL LOCATION:
AC DISCONNECT
NEC: 2020
(NEC 690.13(B))

!

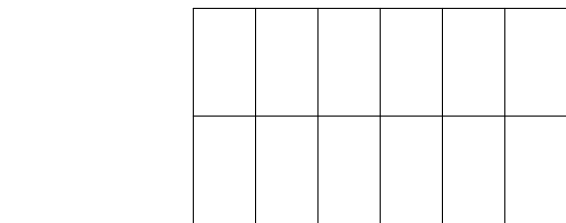
WARNING

PHOTOVOLTAIC SYSTEM COMBINER PANEL

DO NOT ADD LOADS

LABEL LOCATION:
UTILITY SERVICE ENTRANCE/METER, INVERTER/DC DISCONNECT IF REQUIRED BY LOCAL AHJ, OR OTHER LOCATIONS AS REQUIRED BY LOCAL AHJ.
PER CODE(S): NEC 2020: ARTICLE 690.56(C)

PV ARRAY

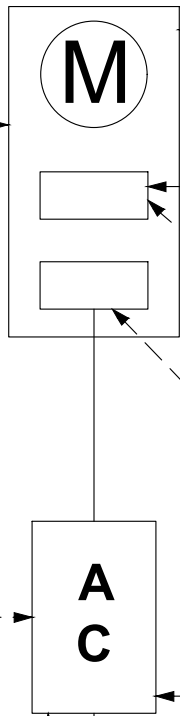


!

WARNING

PHOTOVOLTAIC POWER SOURCE

LABEL LOCATION :
EXPOSED RACEWAY, CABLE TRAYS, JUNCTION BOX
NEC: 2020, PER CODE: NEC 690.31(D)(2)



!

WARNING

POWER SOURCE OUTPUT CONNECTION. DO NOT RELOCATE THIS OVERCURRENT DEVICE.

LABEL LOCATION:
MAIN SERVICE DISCONNECT
NEC: 2020, 705.12(B)(3)(2)

MAIN PHOTOVOLTAIC SYSTEM DISCONNECT

LABEL LOCATION: MAIN SERVICE DISCONNECT, NEC 2020, 690.13(B)

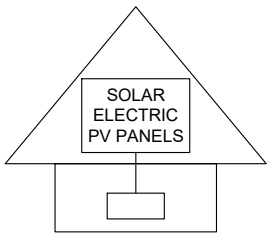
PV SOLAR BREAKER

DO NOT RELOCATE THIS OVERCURRENT DEVICE

LABEL LOCATION: MAIN SERVICE PANEL, NEC 2020, 690.13(B)

SOLAR PV SYSTEM EQUIPPED WITH RAPID SHUTDOWN

TURN RAPID SHUTDOWN SWITCH TO THE "OFF" POSITION TO SHUTDOWN PV SYSTEM AND REDUCE SHOCK HAZARD IN THE ARRAY



AT INVERTER [IFC 605.11.3.1(1) & 690.56(C)]
PER CODE: NEC 2020

SOLAR PHOTOVOLTAIC SYSTEMS

(PER CODE: NEC 690)

SOLAR PHOTOVOLTAIC SYSTEMS

(PER CODE: NEC 690)

!

WARNING

DUAL POWER SUPPLY

SOURCES: UTILITY GRID AND PV SOLAR ELECTRIC SYSTEM

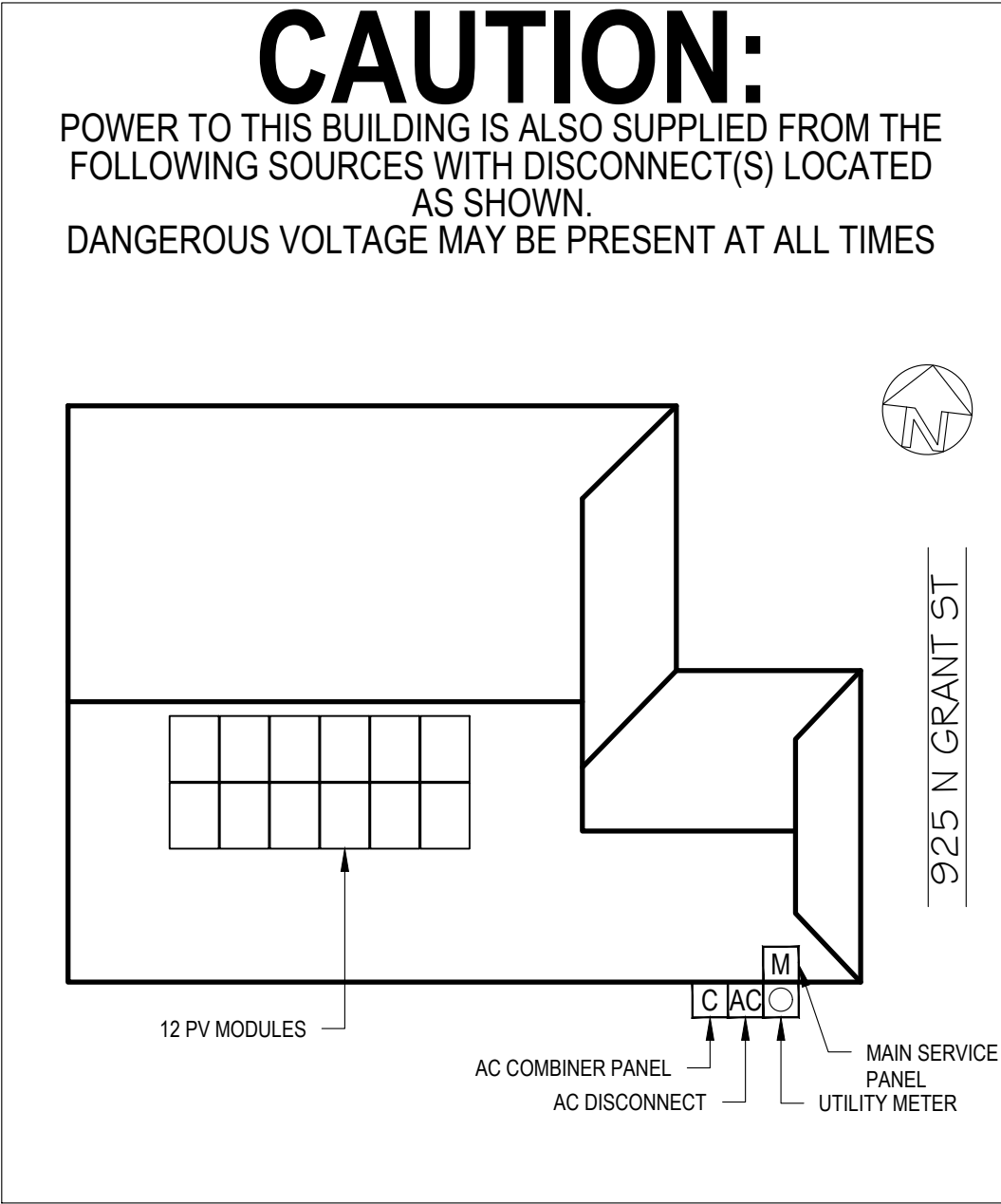
LABEL LOCATION:
PRODUCTION / NET METER (BI-DIRECTIONAL)
NEC: 2020
NEC 690.59, 705.12(D)(3)

PHOTOVOLTAIC AC DISCONNECT

MAXIMUM AC OPERATING CURRENT 19.20 AMPS

NOMINAL OPERATING VOLTAGE 240 VOLTS

LABEL LOCATION:
AC DISCONNECT, POINT OF INTERCONNECTION
NEC: 2020
(PER CODE: NEC 690.54)



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WARNING LABELS

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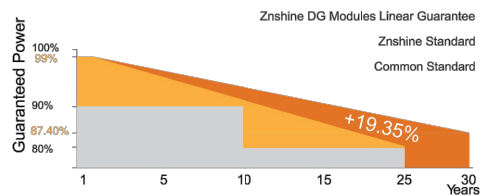
ZXM7-UHLD108 Series

16BB HALF-CELL N-Type TOPCon
Double Glass Monocrystalline PV Module

425-440W POWER RANGE 22.53% MAXIMUM EFFICIENCY 0.40% YEARLY DEGRADATION

12 12 YEARS PRODUCT WARRANTY 30 30 YEARS OUTPUT GUARANTEE

12 years product warranty for general application
15 years product warranty for Rooftop PV system
30 years output warranty / 0.40% Annual Degradation over 30 years



*Please check the valid version of Limited Product Warranty which is officially released by ZNSHINE PV-TECH Co.,Ltd.



IEC 61215/IEC 61730

ISO 14001: Environmental Management System

ISO 9001: Quality Management System

ISO45001: Occupational Health and Safety Management System

*As there are different certification requirements in different markets, please contact your local znshine sales representative for the specific certificates applicable to the products in the region in which the products are to be used.

KEY FEATURES



Excellent Cells Efficiency

SMBB technology reduce the distance between busbars and finger grid line which is benefit to power increase.



Better Weak Illumination Response

More power output in weak light condition, such as haze, cloudy, and early morning.



TIER 1

Global, Tier 1 bankable brand, with independently certified advanced automated manufacturing.



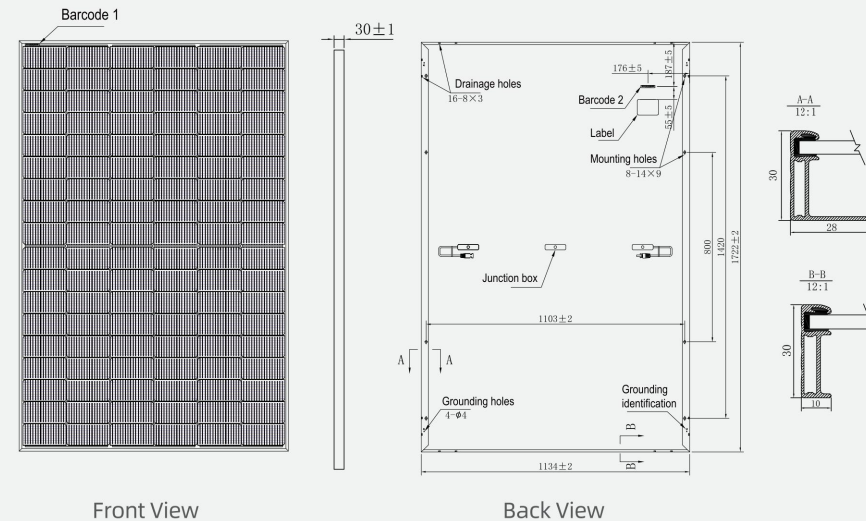
Excellent Quality Management System

Warranted reliability and stringent quality assurances well beyond certified requirements.

ZXM7-UHLD108 Series | Znshinesolar 16BB HALF-CELL N-Type TOPCon Double Glass Monocrystalline PV Module

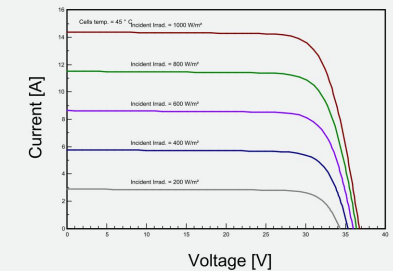


DIMENSIONS OF PV MODULE(mm)

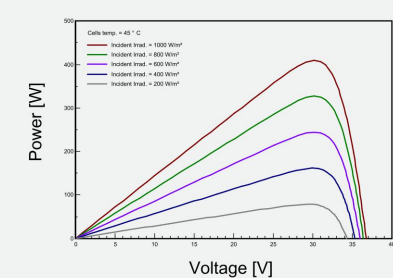


*Remark: customized frame color and cable length available upon request

I-V CURVES OF PV MODULE(435W)



P-V CURVES OF PV MODULE(435W)



ELECTRICAL CHARACTERISTICS | STC*

Module Type	ZXM7 UHLD108-425/N	ZXM7 UHLD108-430/N	ZXM7 UHLD108-435/N	ZXM7 UHLD108-440/N
Nominal Power Watt Pmax(W)*	425±5	430±5	435±5	440±5
Maximum Power Voltage Vmp(V)	31.70	31.90	32.10	32.30
Maximum Power Current Imp(A)	13.41	13.47	13.56	13.63
Open Circuit Voltage Voc(V)	38.30	38.50	38.70	38.90
Short Circuit Current Isc(A)	14.12	14.19	14.26	14.39
Module Efficiency (%)	21.76	22.02	22.28	22.53

*The data above is for reference only and the actual data is in accordance with the practical testing

*STC (Standard Test Condition): Irradiance 1000W/m², Module Temperature 25±2°C, AM 1.5

*Measuring uncertainty: ±3%, all the electrical characteristics such as Power, Im, Vm and FF are within ±3% tolerance.

MECHANICAL DATA

Solar cells	N-type Monocrystalline
Cells orientation	108 (6×18)
Module dimension	1722×1134×30 mm (With Frame)
Weight	24.5±1.0 kg
Glass	2.0 mm+2.0mm, High Transmission, AR Coated Heat Strengthened Glass
Junction box	PV-XT1609Nxyz, IP 68, 3 diodes
Cables	H1Z2Z2-K 1×4,0mm²
Connectors*	PV-XT101.1 Suzhou Xtong Photovoltaic Technologies Co., Ltd. PV Modules manufactured in china

*Please refer to regional datasheet for specified connector

ELECTRICAL CHARACTERISTICS | NMOT

Maximum Power Pmax(Wp)	320.90	324.30	328.60	332.3
Maximum Power Voltage Vmp(V)	29.90	30.10	30.30	30.40
Maximum Power Current Imp(A)	10.73	10.78	10.84	10.93
Open Circuit Voltage Voc(V)	36.10	36.30	36.50	36.70
Short Circuit Current Isc(A)	11.39	11.45	11.51	11.61

*NMOT: Irradiance 800W/m², Ambient Temperature 20°C, AM 1.5, Wind Speed 1m/s

PACKAGING CONFIGURATION*

Piece/Box	36
Piece/Container(40'HQ)	936

*Customized packaging is available upon request.

TEMPERATURE RATINGS

NMOT	44°C ±2°C	Maximum system voltage	1500 V DC
Temperature coefficient of Pmax	(-0.30±0.03)%/°C	Operating temperature	-40°C~+85°C
Temperature coefficient of Voc	-0.25%/°C	Maximum series fuse	25 A
Temperature coefficient of Isc	0.046%/°C	Maximum load front/back	3600/1600 with safety factor 1.5
		Fire safety class	Class A

*Remark: Do not connect Fuse in Combiner Box with two or more strings in parallel connection

*Remark: Electrical data in this catalog do not refer to a single module and they are not part of the offer.

They only serve for comparison among different module types.

*Caution: Please be kindly advised that PV modules should be handled and installed by qualified people who have professional skills

and please carefully read the safety and installation instructions before using our PV modules.

Founded in 1988, Znshine solar is a world's leading high-tech PV module manufacturer. With the advanced production lines, the company boasts module capacity of 10 GW. Bloomberg has listed Znshine as a global Tier 1 PV module maker. Today Znshine has distributed its sales to more than 60 countries around the globe.

www.znshinesolar.com

Address: 1#, Zhixi Industrial Zone, Jintan Jiangsu 213251, P.R. China Tel: +86 519 6822 0233 E-mail: info@znshinesolar.com

Note: Specifications included in this datasheet are subject to change without notice. ZNSHINE reserves the right of final interpretation © ZNSHINE SOLAR 2023 | Version: ZXM7-UHLD108 2307.E

No special undertaking or warranty for the suitability of special purpose or being installed in extraordinary surroundings is granted unless as otherwise specifically committed by manufacturer in contract document



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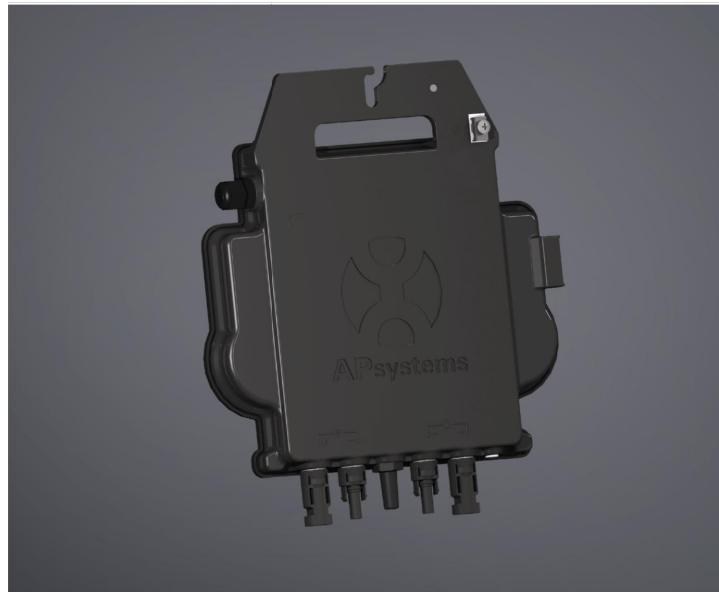
Sheet Name
PV MODULE
SPECIFICATION
SHEET

Sheet Size

ANSI B
11" X 17"

Sheet Number

DS 01



DS3 Series

The most powerful Dual Microinverter

- One microinverter connects to two solar modules
- Max output power reaching 640VA, 768VA or 880VA
- Two independent input channels (MPPT)
- CA Rule 21 (UL 1741 SB) compliant
- NEC 2020 690.12 Rapid Shutdown Compliant
- Encrypted Wireless ZigBee Communication
- Phase Monitored and Phase Balanced

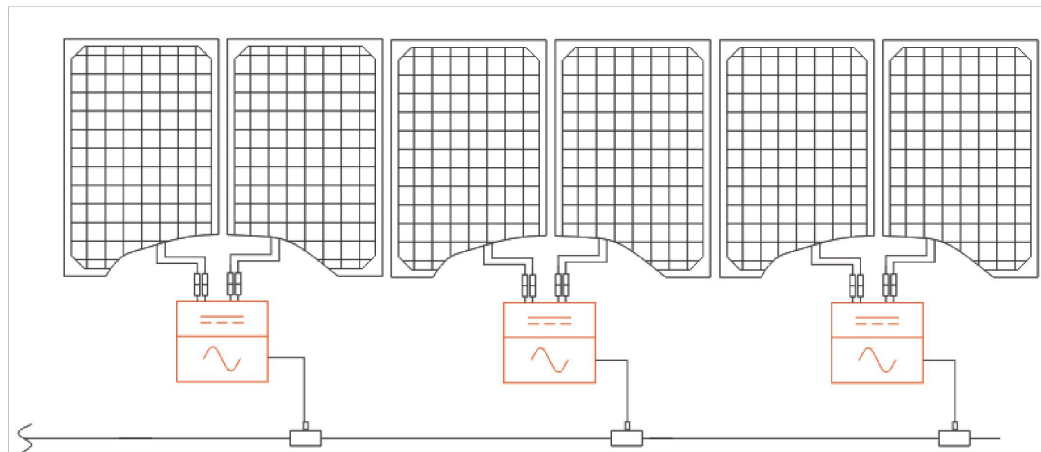
PRODUCT FEATURES

APsystems' 3rd generation of dual-module microinverters, the DS3 product family represents the culmination of years of power conversion expertise and innovation in high-efficiency, high-density power conversion to maximize the peak performance of today's high-capacity PV modules.

The DS3 series reaches unprecedented levels of power output. It features 2 input channels, each with independent MPPT, and encrypted wireless ZigBee communication. An innovative and compact design makes the product lighter while maximizing power production, and silicone-encapsulated components reduce stress on electronics, facilitate thermal dissipation, and enhance weatherproofing. Reliability is significantly increased thanks to 20% fewer components than previous generations. A 24/7 energy access through apps or web based portal facilitate remote diagnosis and maintenance.

The DS3 series is grid-interactive and fully compliant with CA Rule 21 requirements. With an excellent performance and high conversion efficiency, a unique integration with less components, the APsystems DS3 series is a gamechanger for residential and commercial solar.

WIRING SCHEMATIC



2024/02/22 Rev2.0

Datasheet | DS3 Microinverter Series

Model	DS3-S	DS3-L	DS3
Region		USA / Canada	
Input Data (DC)			
Recommended PV Module Power (STC) Range	250Wp-480Wp+	265Wp-570Wp+	300Wp-660Wp+
Peak Power Tracking Voltage		28V-45V	
Operating Voltage Range		26V-60V	
Maximum Input Voltage		60V	
Maximum Input Current	16A x 2	18A x 2	20A x 2
Maximum input short circuit current	20A per input	22.5A per input	25A per input
Output Data (AC)			
Maximum Continuous Output Power	640VA	768VA	880VA
Nominal Output Voltage/Range ⁽¹⁾		240V / 211V-264V	
Nominal Output Current	2.66A	3.2A	3.7A
Maximum Output Fault Current (ac) And Duration	5.691Apk, 26.75ms of duration; 3.307Arms		
Nominal Output Frequency/ Range ⁽¹⁾	60Hz/58.8Hz-61.2Hz(HECO:57Hz-63Hz)		
Power Factor (Default/Adjustable)	0.99/0.8 leading...0.8 lagging		
Maximum Units per 12AWG Branch ⁽²⁾	6 (20A breaker)	5 (20A breaker)	4 (20A breaker)
Maximum Units per 10AWG Branch ⁽²⁾	9 (30A breaker)	7 (30A breaker)	6 (30A breaker)

Efficiency

Peak Efficiency	97.3%
CEC Efficiency	97%
Nominal MPPT Efficiency	99.5%
Night Power Consumption	20mW

Mechanical Data

Operating Ambient Temperature Range ⁽³⁾	-40°F to +149°F (-40°C to +65°C)	
Storage Temperature Range	-40°F to +185°F (-40°C to +85°C)	
Dimensions (W x H x D)	10.3" x 8.6" x 1.6" (263mm x 218mm x 41.2mm)	10.3" x 8.6" x 1.7" (263mm x 218mm x 42.5mm)
Weight	5.7lbs(2.7kg)	6.8lbs(3.1kg)
DC Connector Type	Stäubli MC4 PV-ADBP4-S2&ADSP4-S2	
Cooling	Natural Convection - No Fans	
Enclosure Environmental Rating	Type 6	

Features

Communication (Inverter To ECU) ⁽⁴⁾	Encrypted ZigBee
Isolation Design	High Frequency Transformers, Galvanically Isolated
Energy Management	Energy Management Analysis (EMA) system
Warranty ⁽⁵⁾	10 Years Standard ; 25 Years Optional

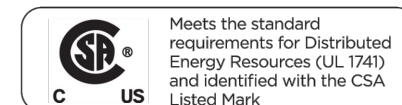
Compliance

Safety and EMC Compliance	UL1741; CSA C22.2 No. 107.1-16; UL1741SA; UL1741SB; IEEE1547; Rule 21; SRD-V2.0; FCC Part15; ICES-003; NEC2014&NEC2017&NEC2020 Section 690.11 DC Arc-Fault circuit Protection; NEC2014&NEC2017&NEC2020 Section 690.12 Rapid Shutdown of PV systems on Buildings
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(1) Nominal voltage/frequency range can be extended beyond nominal if required by the utility.
(2) Limits may vary. Refer to local requirements to define the number of microinverters per branch in your area.
(3) The inverter may enter to power de-grade mode under poor ventilation and heat dissipation installation environment.
(4) Recommend no more than 80 inverters register to one ECU for stable communication.
(5) To be eligible for the warranty, APsystems microinverters need to be monitored via the EMA portal. Please refer to our warranty T&Cs available on usa.APsystems.com.

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Specifications subject to change without notice please ensure you are using the most recent update found at web : usa.APsystems.com

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8627 N. Mopac Expy, Suite 150, Austin, TX 78759
apsystems.com



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Project Name &
Address

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Sheet Name
**MICROINVERTER
SPECIFICATION
SHEET**

Sheet Size

**ANSI B
11" X 17"**

Sheet Number

DS 02



Powering Business Worldwide

pe.eaton.com

BR style 1-inch loadcenter

BR816L125RP

UPC:786676001472

Dimensions:

- Height: 13 IN
- Length: 3.56 IN
- Width: 11 IN

Warranties:

- 10 year

Specifications:

- **Special Features:** Current design
- **Type:** Main lug
- **Amperage Rating:** 125A
- **Box Size:** 7r
- **Bus Material:** Aluminum
- **Cover:** Cover included
- **Enclosure:** NEMA 3R
- **Enclosure Material:** Metallic
- **Interrupt Rating:** 10 kAIC
- **Main Circuit Breaker:** BR
- **Number Of Circuits:** 16
- **Number Of Spaces:** 8
- **Number Of Wires:** Three-wire
- **Phase:** Single-phase
- **Voltage Rating:** 120/240V
- **Wire Size:** #14-1 AWG Cu/Al

Supporting documents:

- [Eatons Volume 1-Residential and Light Commercial](#)
- [Cutler-Hammer Type CH and BR Loadcenters - Instructions](#)
- [Type BR Arc Fault Circuit Breakers and Loadcenters](#)
- [Eaton Specification Sheet - BR816L125RP](#)



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COMBINER PANEL
CERTIFICATION

Sheet Size

ANSI B
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Sheet Number

DS 03

1.1

Switching Devices

Safety Switches

1

Product Overview

Product Selection Guide

Safety Switch

Switch Type	NEMA Rating	Fused/non-fused	Number of Poles				
			1	2	3	4	6
Heavy-duty single-throw	NEMA 1	Fused and non-fused ①	—	30–800 A	30–1200 A	30–600 A	—
	NEMA 3R	Fused and non-fused ②	30–600 A	30–800 A	30–1200 A	—	—
	NEMA 12/3R	Fused and non-fused ①②	30–600 A	30–800 A	30–1200 A	30–600 A	30–200 A
	NEMA 4	Fused and non-fused	—	400–800 A	400–800 A	—	—
	NEMA 7/9	Fused and non-fused ③	—	—	30–100 A	—	—
	NEMA 4X non-metallic	Fused and non-fused ③	—	—	30–200 A	—	—
	NEMA 4X stainless steel (304 or 316 Grade)	Fused and non-fused ①③	30–600 A	30–800 A	30–1200 A	30–600 A	30–200 A
General-duty single-throw	NEMA 1 and 3R	Fused	—	30–600 A	30–600 A	—	—
	NEMA 1	Non-fused	—	30–100 A	30–600 A	—	—
	NEMA 3R	Non-fused	—	30–200 A	30–600 A	—	—
Heavy-duty double-throw	NEMA 1	Fused	—	200 A	30–1200 A	—	—
	NEMA 3R	Fused ④	—	200 A	100–1200 A	—	—
	NEMA 12	Fused	—	—	30–400 A	—	—
	NEMA 4X stainless steel (304 or 316 Grade)	Fused	—	—	30–400 A	—	—
	NEMA 1	Non-fused	—	30–1200 A	30–1200 A	200–800 A	—
	NEMA 3R	Non-fused ④	—	200–1200 A	30–1200 A	30–800 A	30–100 A
	NEMA 12	Non-fused	—	—	30–800 A	—	—
	NEMA 4X stainless steel (304 or 316 Grade)	Non-fused	—	—	30–800 A	—	—
General-duty double-throw	NEMA 3R	Non-fused	—	30–400 A	—	—	—
Shunt trip safety switches	NEMA 12/3R, 4, 4X stainless steel (304 or 316 stainless steel)	Fused and non-fused	—	30–1200 A	30–1200 A	30–400 A	—
Window switches	NEMA 12/3R and 4X stainless steel (304 or 316 Grade)	Fused and non-fused ①	—	30–1200 A	30–1200 A	30–600 A	30–200 A
EnviroLine stainless steel switch	NEMA 4X (304 or 316-Grade)	Fused and non-fused ⑤	—	30–400 A	30–400 A	—	—
Mill duty switches	NEMA 12/3R and 4X stainless steel (304 or 316 Grade)	Fused and non-fused	—	30–600 A	30–600 A	—	—
Quick connect double-throw	NEMA 1 and 3R	Fused and non-fused ④	—	100–1200 A	100–1200 A	100–1200 A	—
Quick connect single-throw	NEMA 1 and 3R	Fused and non-fused ④	—	100–1200 A	100–1200 A	100–1200 A	—
Elevator control switches	NEMA 1, 3R, 4, and 12	Fused	—	—	30–400 A	—	—
OEM Line Isolation Switches (OLI)	NEMA 12/3R and 4X stainless steel (304 or 316 Grade)	Fused and non-fused	—	30–400 A	30–400 A	—	—
Receptacle switches	NEMA 12/3R and 4X stainless steel (304 or 316 Grade)	Fused and non-fused ⑥	—	—	30–100 A	—	—
Auxiliary power switch	NEMA 3R	Fused and non-fused ④	—	—	30–200 A	—	—
Left-handed switches	NEMA 1, 3R, 12, and 4X stainless steel (304 or 316 Grade)	Fused and non-fused ⑦	—	—	30–200 A	—	—
Surge switches	NEMA 12/3R and 4X stainless steel (304 or 316 Grade)	Fused and non-fused	—	30–1200 A	30–1200 A	—	—
Heavy-duty double-door	NEMA 12/3R and 4X stainless steel (304 or 316 Grade)	Fused	—	30–1200 A	30–1200 A	—	—
Enclosed rotary	NEMA 1, 12/3R, NEMA 4X stainless steel	Non-fused	—	—	16–80 A	16–80 A	—
	NEMA 4X non-metallic	Non-fused	—	—	16–80 A	16–80 A	—
	NEMA 4X polycarbonate non-metallic	Non-fused	—	—	30 A and 60 A	16–80 A	—

- Notes
- ① 400–800 A 6-pole, and 800 A four-pole heavy-duty, single-throw switches are non-UL Listed and available.

② No viewing windows or other enclosure modifications are available with this product.

③ EnviroLine stainless steel switches have stainless steel enclosures, sub pans, handles, and mechanisms.

④ All left-handed switches are non-UL Listed. Higher amperages are not available.

⑤ Single-pole product is the single circuit 600 Vdc switch series.

⑥ NEMA 3R stainless versions are available and are UL Listed.

⑦ Switches with non-standard receptacles or at higher amperages are non-UL Listed and available.

Cross-Reference

General-Duty

Ampere Rating	Catalog Number Eaton	General Electric	Siemens	Square D
Fusible, Two-Pole, Three-Wire, 240 Vac, NEMA 1				
30	DG221NGB	TG3221	GF221N	D221N
60	DG222NGB	TG3222	GF222N	D222N
100	DG223NGB	TG3223	GF223N	D223N
200	DG224NGK	TG3224	GF224N	D224N
400	DG225NGK	TG3225	GF225N	D225N
600	DG226NGK	TG3226	GF226N	D226N
Fusible, Three-Pole, Three-Wire, 240 Vac, NEMA 1				
30	DG321FGB	TG4321	GF321N	D321N
60	DG322FGB	TG4322	GF322N	D322N
100	DG323FGB	TG4323	GF323N	D323N
200	DG324FGK	TG4324	GF324N	D324N
400	DG325FGK	TG3325	GF325N	D325N
600	DG326FGK	TG3326	GF326N	D326N
Fusible, Three-Pole, Four-Wire, 240 Vac, NEMA 1				
30	DG321NGB	TG4321	GF321N	D321N
60	DG322NGB	TG4322	GF322N	D322N
100	DG323NGB	TG4323	GF323N	D323N
200	DG324NGK	TG4324	GF324N	D324N
400	DG325NGK	TG4325	GF325N	D325N
600	DG326NGK	TG4326	GF326N	D326N
Non-Fusible, Two-Pole, Two-Wire, 240 Vac, NEMA 1				
30	DG221UGB	TGN3321	N/A	N/A
60	DG222UGB	TGN3322	N/A	Q0260NATS
100	DG223UGB	TGN3323	N/A	Q02000NS
200	DG324UGK	TGN3324	N/A	DU324
400	DG325UGK	TGN3325	N/A	DU325
600	DG326UGK	TGN3326	N/A	DU326
Non-Fusible, Three-Pole, Three-Wire, 240 Vac, NEMA 1				
30	DG321UGB	TGN3321	GNF321	DU321
60	DG322UGB	TGN3322	GNF322	DU322
100	DG323UGB	TGN3323	GNF323	DU323
200	DG324UGK	TGN3324	GNF324	DU324
400	DG325UGK	TGN3325	GNF325	DU325
600	DG326UGK	TGN3326	GNF326	DU326

Switching Devices

Safety Switches

1.1

1

General-Duty, continued

Ampere Rating	Catalog Number Eaton	General Electric	Siemens	Square D
Fusible, Two-Pole, Three-Wire, 240 Vac, NEMA 3R				
30	DG221NRB	TG3221R	GF221NR	D221NRB
60	DG222NRB	TG3222R	GF222NR	D222NRB
100	DG223NRB	TG3223R	GF223NR	D223NRB
200	DG224NRK	TG3224R	GF224NR	D224NRB
400	DG225NRK	TG3225R	GF225NR	D225NR
600	DG226NRK	TG3226R	GF226NR	D226NR
Fusible, Three-Pole, Three-Wire, 240 Vac, NEMA 3R				
30	DG321FRB	TG4321R	GF321NR	D321NRB
60	DG322FRB	TG4322R	GF322NR	D322NRB
100	DG323FRB	TG4323R	GF323NR	D323NRB
200	DG324FRK	TG4324R	GF324NR	D324NRB
400	DG325FRK	TG3325R	GF325NR	D325NR
600	DG326FRK	TG3326R	GF326NR	D326NR
Fusible, Three-Pole, Four-Wire, 240 Vac, NEMA 3R				
30	DG321NRB	TG4321R	GF321NR	D321NRB
60	DG322NRB	TG4322R	GF322NR	D322NRB
100	DG323NRB	TG4323R	GF323NR	D323NRB
200	DG324NRK	TG4324R	GF324NR	D324NRB
400	DG325NRK	TG3325R ①	GF325NR	D325NR
600	DG326NRK	TG3326R ①	GF326NR	D326NR
Non-Fusible, Two-Pole, Two-Wire, 240 Vac, NEMA 3R				
30	DG221URB	TGN3321R	GNF321R	DU221RB
60	DG222URB	TGN3322R	GNF322R	DU222RB
100	DG223URB	TGN3323R	GNF323R	Q02000NRB
200	DG324URK	TGN3324R	GNF324R	DU324RB
Non-Fusible, Three-Pole, Three-Wire, 240 Vac, NEMA 3R				
30	DG321URB	TGN3321R	GNF321R	DU321RB
60	DG322URB	TGN3322R	GNF322R	DU322RB
100	DG323URB	TGN3323R	GNF323R	DU323RB
200	DG324URK	TGN3324R	GNF324R	DU324RB
400	DG325URK	N/A	N/A	N/A
600	DG326URK	N/A	N/A	N/A

- Notes
- ① Separate neutral kit required.

Always verify the number of poles and wires required since catalog numbers may appear in multiple tables.



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INFINITY.SOLAR.COM

REVISIONS

Description	Date	Rev
Initial Design	6/16/26	00

Signature with Seal

Project Name & Address

KEVIN GONZALEZ RESIDENCE

925 N GRANT ST
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Sheet Name

AC DISCONNECT
UL CERTIFICATION

Sheet Size

ANSI B
11" X 17"

Sheet Number

DS 04

RSTC Enterprises, Inc.
2214 Heimstead Road
Eau Claire, WI 54703
715-830-9997



Outdoor Photovoltaic Enclosures

Composition/Cedar Roof System

ETL listed and labeled

Report # 3171411PRT-002 Revised May, 2018

- UL50 Type 3R, 11 Edition Electrical equipment enclosures
- CSA C22.2 No. 290 Nema Type 3R
- Conforms to UL 1741 Standard

0799 Series Includes:

- 0799 - 2 Wire size 2/0-14
- 0799 - 5 Wire size 14-6
- 0799 - D Wire size 14-8

Models available in Grey, Black or Stainless Steel

Basic Specifications

- Material options:
- Powder coated, 18 gauge galvanized 90 steel (1,100 hours salt spray)
 - Stainless steel

Process - Seamless draw (stamped)
Flashing - 15.25" x 17.25"
Height - 3"
Cavity - 255 Cubic inches

Base Plate:

- Fastened to base using toggle fastening system
- 5 roof deck knockouts
- Knockout sizes: (3) .5", (1) .75" and (1) 1"
- 8", 35mm slotted din rail
- Ground Block

Passthrough and combiner kits are available for either
AC or DC applications.

0799 Series




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Sheet Name

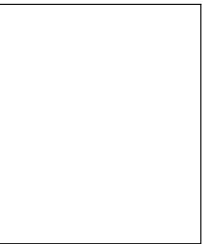
JUNCTION BOX UL
CERTIFICATION

Sheet Size

ANSI B
11" X 17"

Sheet Number

DS 05



Dual Pass 2kV Rated PV



Bare Copper

XLPE Insulation

PVC Jacket



Cable Identification

“ADVANCED DIGITAL CABLE INC. XX AWG (UL)
PV WIRE 2KV 90C WET OR 105C DRY SUN RES
UV RATED VW-1 OR RHW-2 DIRECT BURIAL RoHS
E324841”

Cross-Linked Polyethylene Insulated • PVC Jacketed
14 - 750 MCM • 2000 Volts • 105°C Dry and 90°C Wet

Description

ADC's **Solarlink** brand Photovoltaic cable has a chemically cross-linked polyethylene insulation with a sunlight resistant PVC jacket.

Applications

Appropriate for use in solar power applications that require 2,000 volt rating. For use in grounded interconnection and un-grounded Photovoltaic power systems.

Construction

Conductors: Stranded bare copper conductor per ASTM B-3, B-8. Available in 7 or 19 stranded versions as well as tinned copper.

Insulation: White Chemically Cross-linked polyethylene with colored sunlight resistant PVC jacket.

Colors: Black, Green, White, Red. Print on one side with a contrasting ink. An extruded stripe and other colors are available upon request.

Industry Listings & Standards

UL Listed as PV per UL Standard 4703
RHW-2 per UL Standard 44 and USE-2 per UL Standard 854
90°C Wet/105°C Dry Rated
Gasoline and Oil Resistant II
RoHS Compliant
Sunlight Resistant
VW-1 Rated



Cable Data

Part Number	AWG	Strand	Insulation Thickness (mils)	Jacket Thickness (mils)	Nominal O.D. (inch)	Approximate Net Weight lbs/1M'
3142DPV	14	7	60	30	.251	39
3122DPV	12	7	60	30	.270	49
3102DPV	10	7	60	30	.294	65
3082DPV	8	7	70	30	.344	95
3062DPV	6	7	70	30	.381	130
3042DPV	4	7	70	30	.429	188
3032DPV	3	7	70	30	.457	228
3022DPV	2	7	70	30	.489	277
3012DPV	1	19	90	30	.569	358
30102DPV	1/0	19	90	30	.602	427
30202DPV	2/0	19	90	30	.646	519
30302DPV	3/0	19	90	30	.696	633
30402DPV	4/0	19	90	30	.752	784
302502DPV	250 MCM	37	105	30	.828	934
303002DPV	300 MCM	37	105	30	.875	1104
303502DPV	350 MCM	37	105	30	.931	1260
304002DPV	400 MCM	37	105	30	.976	1431
305002DPV	500 MCM	37	105	30	1.059	1745
306002DPV	600 MCM	61	120	30	1.166	2097
307502DPV	750 MCM	61	120	30	1.268	2612

The information contained on this specification is intended to be used as a guide in product selection and is believed to be reliable. ADC has made every effort to ensure the data shown above is accurate at the time of publication. This specification is subject to change anytime without notice. REV0821

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Description	Date	Rev
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Sheet Name

WIRING
SPECIFICATION

Sheet Size

ANSI B
11" X 17"

Sheet Number

DS 06



RAD MOUNT (Rafter And Decking)

Integrated with a L-Foot & Flash with the flexibility to mount to Either the Rafter or Decking. This simple but RADICAL design is made for asphalt shingle roofs. Installers love the ease and faster install times!

PRODUCT LINE

Item

CK-FTS-167RT2

CK-FTS-167BRT2

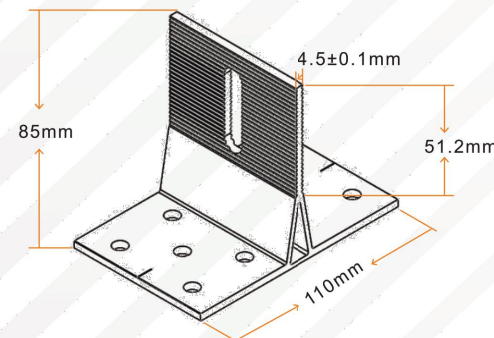
Product Name

RAD Mount Al Roof Hook #167-SILVER

RAD Mount Al Roof Hook #167-BLACK

TECHNICAL DATA

Main Material : AL 6005-T5
Uplift P : N*Fv (N=3)
Wind Velocity : Up to 60 M/S (134 MPH)
Snow Load : 1.4 KN/m²/ 30 lbs/sq. foot
Spacing : Up to 2000mm; 6.5 feet; 80 Inches
Install Site Type : Asphalt Shingle



ORDERING SPECIFICS

Standard Packaging : 25PCS/CTN
Dimensions : 45x27x30/CM
Weight : 21KG

ADVANTAGES

- Optional Holes provided for Rafter & Decking
- Flexible moving on orientations Vertical & Horizontal
- Water-proof with thermoplastic butyl adhesive
- Faster Install Time

COMPONENT LIST

Material	QTY
Rad Mount #167 Hook	01
SUS304 T2 Bolt M8*30	01
SUS304 Flange nut M8	01
Wooden Screw M6.3*65	02
Butyl rubber	01

For Deck Mount; Add Qty 2 Add'l Screws

WARRANTY



INTERTEK LISTED



www.chikousa.com



101 East Baseline Road
Buckeye, AZ 85326

Tel : 1-800-948-5390

Email : info@chikousa.com

CHIKOUSA
RACKING MADE SIMPLE



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Signature with Seal

Project Name &
Address

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Sheet Name

MOUNTING -
DATA SHEET

Sheet Size

ANSI B
11" X 17"

Sheet Number

DS 07

REVISIONS

Description	Date	Rev
Initial Design	6/16/26	00

Signature with Seal

Project Name &
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Sheet Name

RACKING -
DATA SHEET

Sheet Size

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11" X 17"

Sheet Number

DS 08



#518 RAIL

CHIKO 518R aluminum rail is designed for roof mounting system, it could be applied on all roof mount systems. A variety of lengths can help to reduce unnecessary cut.

ADVANTAGES

- Easy installation
- High class anodized
- Universal on roof mount system

WARRANTY



PRODUCT LINE

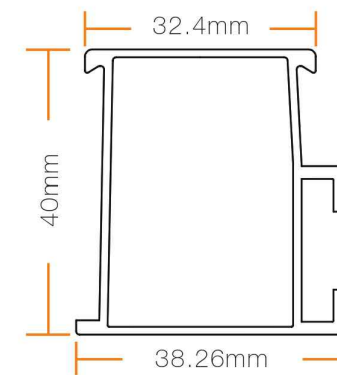
Item	Product Name
CK-FT-R518B1-2100	CHIKO 518 Rail 2100mm
CK-FT-R518B1-3200	CHIKO 518 Rail 3200mm
CK-FT-R518B1-4200	CHIKO 518 Rail 4200mm
CK-FT-R518B1-4350	CHIKO 518 Rail 4350mm

TECHNICAL DATA

Main Material	AL 6005-T5
Wind Velocity	Up to 60 M/S

$I_x=34466.83 \text{ mm}^4$

$I_y=49849.13 \text{ mm}^4$



COMPONENT LIST

MATERIAL	QTY
Aluminium Rail	01

ORDERING SPECIFICS

Standard Packaging	8PCS/PKG
Dimensions	2100/3200/4200/4350mm
Weight	9.6/14.6/19.2/19.85kg



43 RAIL SPLICE KIT

CHIKO 43R aluminium rail splice kit is designed for 43R rail connection from back to position. The most simple and handy installation way.

ADVANTAGES

- Easy installation
- Highclass anodized

WARRANTY

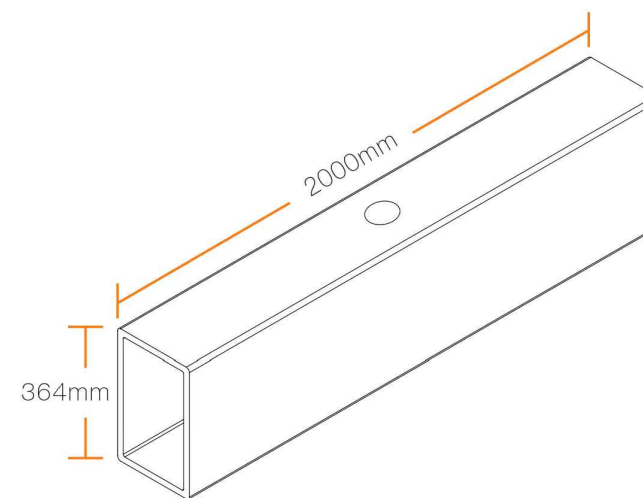


PRODUCT LINE

Item	Product Name
CK-FT-SKA	CHIKO 43 Rail Splice Kit

TECHNICAL DATA

Main Material	AL 6005-T5
Wind Load	Up to 60 M/S
Snow Load	1.4 KM/M ²



COMPONENT LIST

MATERIAL	QTY
Aluminium Rail Splice Kit	01
SUS304 ϕ 4.2*16	04

ORDERING SPECIFICS

Standard Packaging	150 PCS/PKG
Dimensions	51X38X22CM
Weight	30KG



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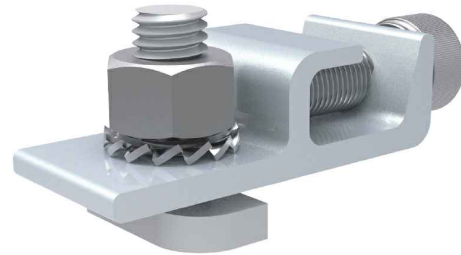
SPLICE KIT
-DATA SHEET

Sheet Size

ANSI B
11" X 17"

Sheet Number

DS 09



Grounding Lug

CHIKO grounding lug is designed for fixing grounding cable going through smoothly between each rails

ADVANTAGES

- Easy installation
- High class anodized
- Tilt- in nut

WARRANTY

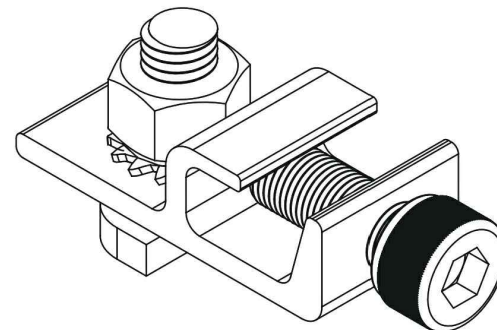


TECHNICAL DATA

Main Material	AL 6005-T5
Tighten torque	15N.m
Safe torque	20N.m

COMPONENT LIST

MATERIAL	QTY
Grounding Lug	01
SUS304 inner hex bots M8X20	01
SUS304 T4 bolts M8x25	01
SUS304 hex nut M8	01
SUS304 spring washer M8	01
SUS304 star washer	01



ORDERING SPECIFICS

Standard Packaging	140 PCS/BOX 560PCS/CTN
Dimensions	51X38X22CM
Weight	5.8/23.2KGS



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Sheet Name

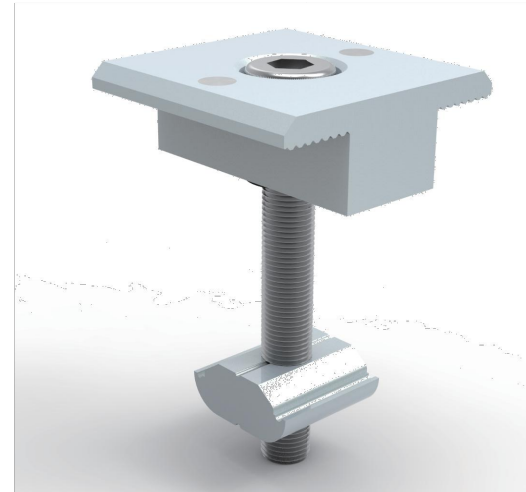
GROUNDING LUG
-DATA SHEET

Sheet Size

ANSI B
11" X 17"

Sheet Number

DS 10



Fix Mid Clamp

CHIKO mid clamps is designed base on 7R rail to fix module between two module, 30mm to 55mm thickness module are available.

ADVANTAGES

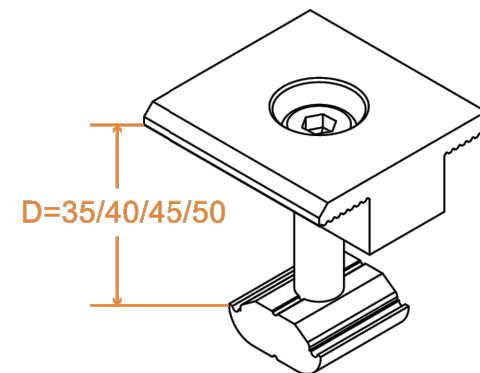
- Easy installation
- High class anodized
- Tilt- in nut

PRODUCT LINE

Item	Product Name
CK-FTM-35	CHIKO Fix Mid Clamp 35mm
CK-FTM-40	CHIKO Fix Mid Clamp 40mm
CK-FTM-45	CHIKO Fix Mid Clamp 45mm
CK-FTM-50	CHIKO Fix Mid Clamp 50mm

PRODUCT LINE

Main Material	AL 6005-T5
Tighten torque	15N.m
Safe torque	20N.m



COMPONENT LIST

MATERIAL	QTY
Mid Clamp	01
SUS304 Bolt M8	01
057 Alu Nut	01

WARRANTY



ORDERING SPECIFICS

Standard Packaging	100 PCS/BOX 400PCS/CTN
Dimensions	51X38X22CM
Weight	26.8/27.5/28/28.6KG



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Sheet Name

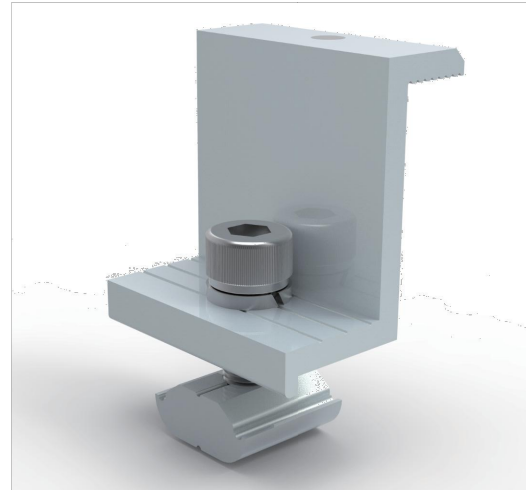
MID CLAMP -
DATA SHEET

Sheet Size

ANSI B
11" X 17"

Sheet Number

DS 11



Fix End Clamp

CHIKO end clamps is designed base on 7R rail to fix module on the end of rail, 30mm to 55mm thickness module are available.

ADVANTAGES

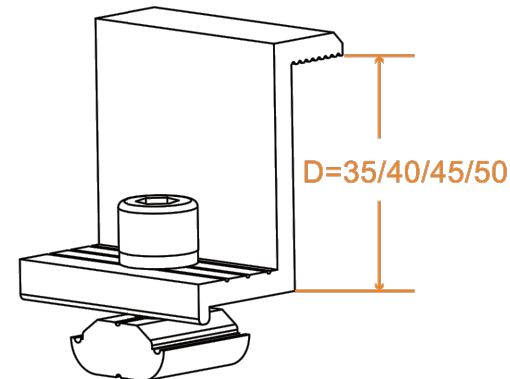
- Easy installation
- High class anodized
- Tilt- in nut

PRODUCT LINE

Item	Product Name
CK-FTE-35	CHIKO Fix End Clamp 35mm
CK-FTE-40	CHIKO Fix End Clamp 40mm
CK-FTE-45	CHIKO Fix End Clamp 45mm
CK-FTE-50	CHIKO Fix End Clamp 50mm

PRODUCT LINE

Main Material	AL 6005-T5
Tighten torque	15N.m
Safe torque	20N.m



COMPONENT LIST

MATERIAL	QTY
End Clamp	01
SUS304 Bolt M8*25	01
SUS304 Washer M8	01
057 Alu Nut	01

WARRANTY



ORDERING SPECIFICS

Standard Packaging	100 PCS/BOX 400PCS/CTN
Dimensions	51X38X22CM
Weight	22/24.5/25.5/26KG



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Sheet Name

END CLAMP -
DATA SHEET

Sheet Size

ANSI B
11" X 17"

Sheet Number

DS 12

This authorizes the application of the Certification Mark(s) shown below to the models described in the Product(s) Covered section when made in accordance with the conditions set forth in the Certification Agreement and Listing Report. This authorization also applies to multiple listee model(s) identified on the correlation page of the Listing Report.
This document is the property of Intertek Testing Services and is not transferable. The certification mark(s) may be applied only at the location of the Party Authorized To Apply Mark.

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Manufacturer: SHANGHAI CHIKO SOLAR TECHNOLOGY CO., LTD
Address: NO.680 Xing Wen Rd.Jiading District, Jiading District Shanghai 201811
Country: China
Party Authorized To Apply Mark: Same as Manufacturer
Report Issuing Office: Intertek Testing Services NA, Inc., Lake Forest, CA
Control Number: 5010190
Authorized by: [Signature] for L. Matthew Snyder, Certification Manager



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Table with 2 columns: Field, Value. Rows include Standard(s), Product, Brand Name, and Models.

infinity solar
INFINITY HOME SOLUTIONS
6405 E MILL PLAIN
VANCOUVER WA 98661
PHONE: 1-800-818-0598
INFINITY SOLAR.COM
REVISIONS
Description Date Rev
Initial Design 6/16/26 00
Signature with Seal
Project Name & Address
KEVIN GONZALEZ RESIDENCE
925 N GRANT ST
LAFAYETTE, OR 97127
PHONE:- N/A
EMAIL:- N/A
Sheet Name
CHIKO USA UL LISTING CERTIFICATE
Sheet Size
ANSI B 11" X 17"
Sheet Number
DS 13

Customer Address:
Customers Name:

Date:
Prepared by:

Identify all fall hazards 10 feet or more above the ground:

- ☐ Opening
- ☐ Skylights
- ☐ Hole
- ☐ Mobile lift work
- ☐ Roof

Methods of fall protection to be used (LSO = Low Slopes Only):

- ☐ Personal fall arrest system
- ☐ Safety watch system (LSO)
- ☐ Warning line system (LSO)
- ☐ Personal fall restraint system
- ☐ Warning line with safety monitor (LSO)

Overhead Hazard Protection Methods:

- ☐ Hard Hats
- ☐ Barricade to control access to area

Nearest Hospital/urgent care:

Manger:

Red dashed lines: Roof edges (fall hazards).

Blue arrows: Ladder/scaffold access points.

Yellow zones: Material storage areas.

Lightning bolt icons: Electrical panels/PV arrays.

Green star: Emergency assembly point (driveway).

Describe Procedures for handling, storage, and securing tools:

Signatures:

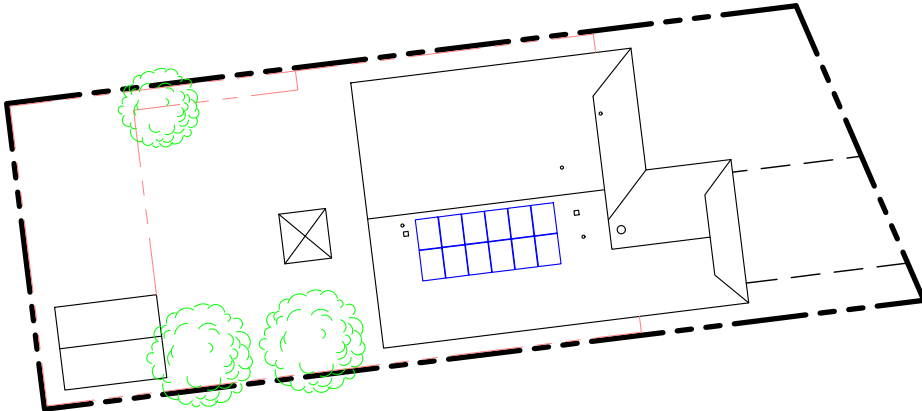
Journeyman Electrician:

Apprentice:

Roof Lead:

Roof Installer:

Roof Installer:



Describe Method for Prompt rescue of employees in the event of a fall:

REVISIONS		
Description	Date	Rev
Initial Design	6/16/26	00

Signature with Seal

Project Name & Address

KEVIN GONZALEZ RESIDENCE
925 N GRANT ST
LAFAYETTE, OR 97127
PHONE:- N/A
EMAIL:- N/A

Sheet Name
SITE SAFETY
MAP

Sheet Size
ANSI B
11" X 17"

Sheet Number
DS 14